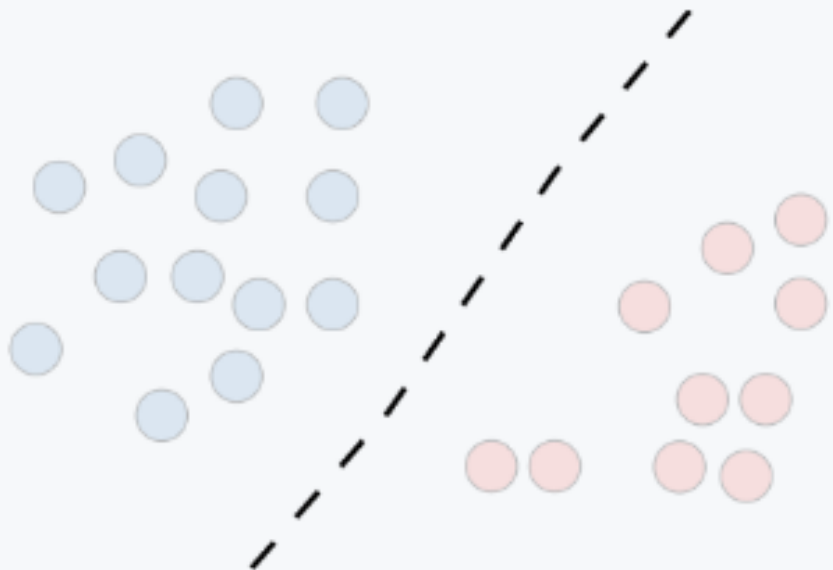
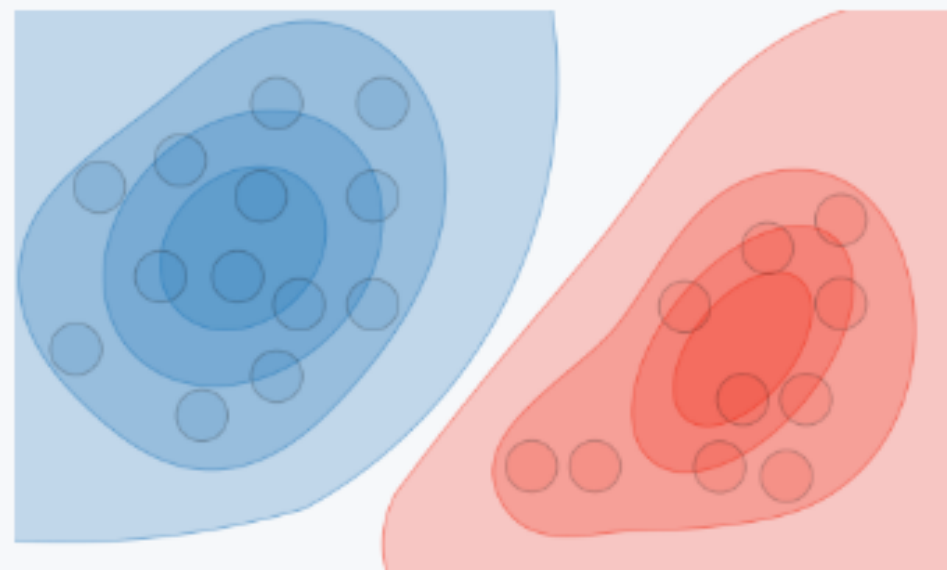




第10章 深度生成模型

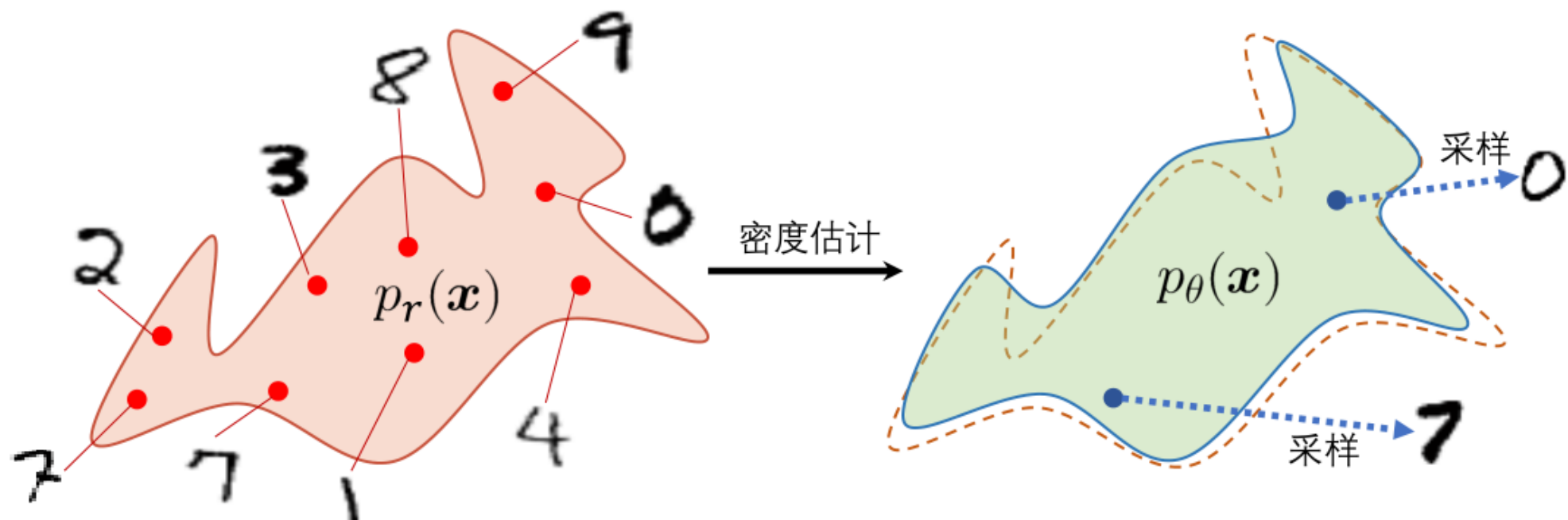
生成模型

机器学习的两种范式

	Discriminative model	Generative model
Goal	Directly estimate $P(y x)$	Estimate $P(x y)$ to then deduce $P(y x)$
What's learned	Decision boundary	Probability distributions of the data
Illustration		
Examples	Regressions, SVMs	GDA, Naive Bayes

生成模型

- 生成模型是一系列用于随机生成可观测数据的模型
- 生成模型包含两个步骤：
 - 密度估计
 - 采样



显式密度模型和隐式密度模型

• 显式密度模型

- 显式地构建出样本的密度函数 $p(x|\theta)$ ，并通过最大似然估计来求解参数；
- 变分自编码器、深度信念网络

• 隐式密度模型

- 不显式地估计出数据分布的密度函数
- 但能生成符合数据分布 $p_{\text{data}}(x)$ 的样本
- 无法用最大似然估计

深度生成模型

- **深度生成模型就是利用神经网络构建的生成模型。**

- 生成对抗网络 (Generative Adversarial Network, GAN)
 - [Goodfellow et al., 2014]
- 变分自编码器 (Variational Autoencoder, VAE)
 - [Kingma and Welling, 2013, Rezende et al., 2014]
- 基于流的方法 (Flow-based Method)
- 扩散模型 (Diffusion Model)

无监督式GAN

生成对抗网络 (GAN)

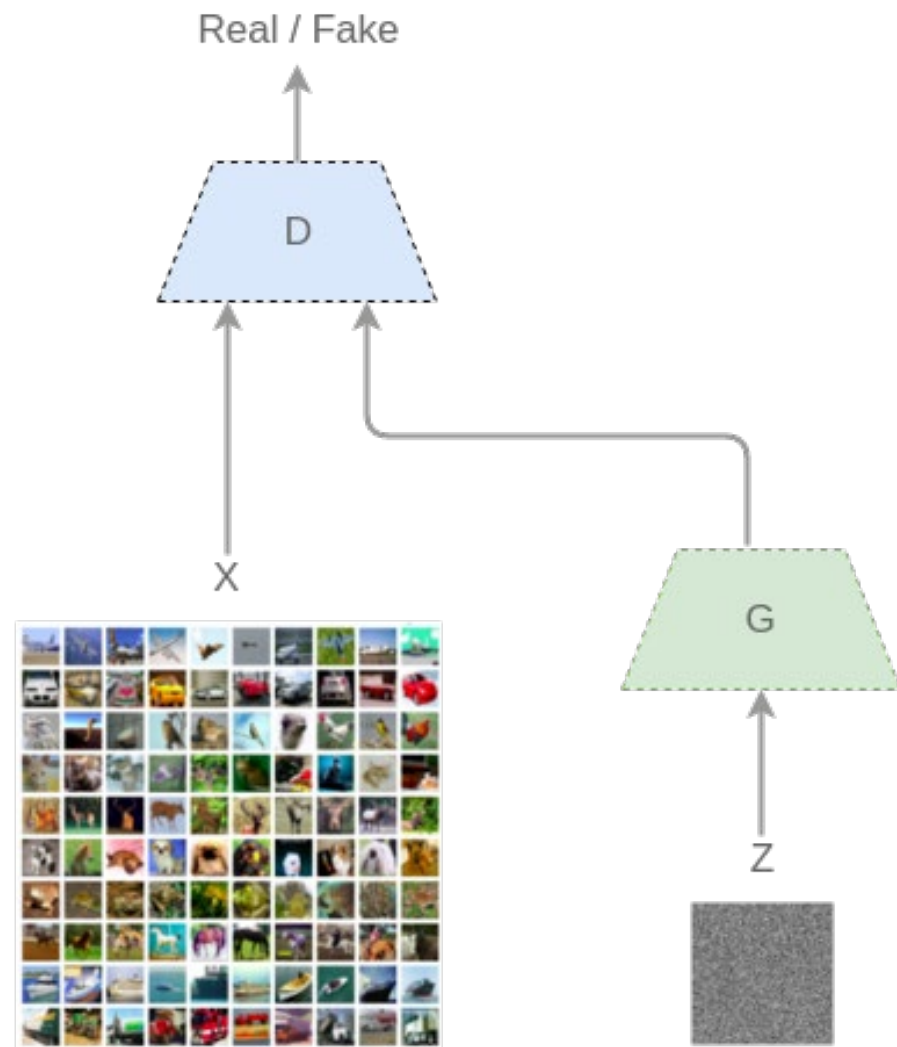
• 生成对抗网络 (Generative Adversarial Network, GAN)

• 生成器 G

- 生成让 D 无法判断真假的样本
- 输入 z : 从一个先验概率分布取样的随机向量
- 输出 $G(z)$: 生成的样本

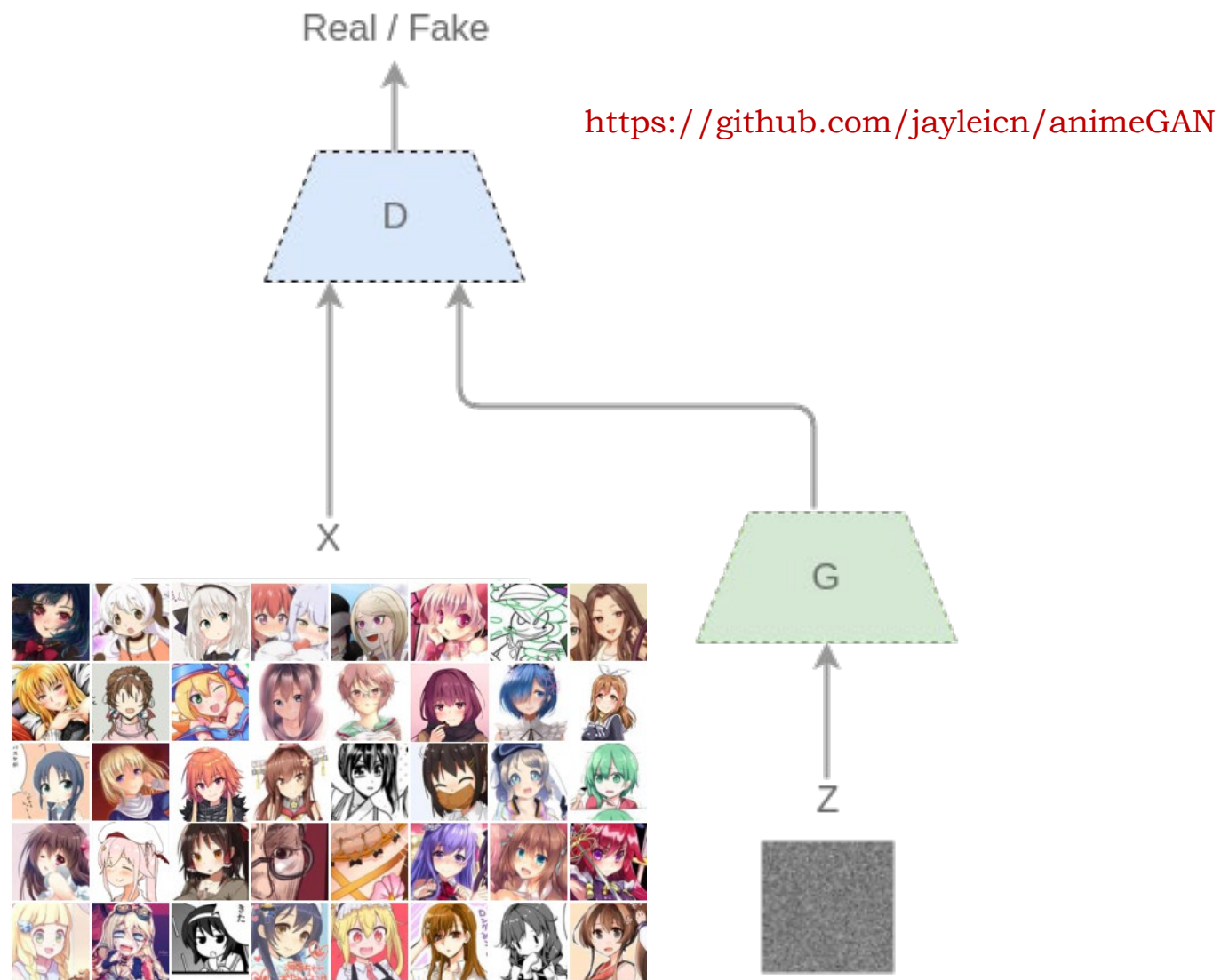
• 判别器 D

- 判断样本是真实的还是生成的
- 输入: 真实样本/G 生成的样本
- 输出: 对真实样本输出 1, 生成样本输出 0

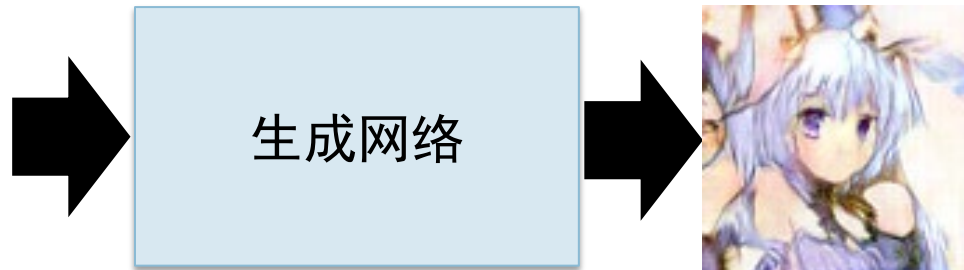


无监督式生成对抗网络

• 示例：生成小姐姐



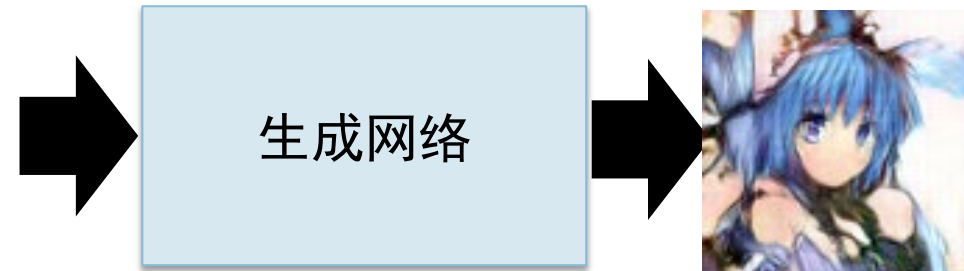
生成网络示例

$$\begin{bmatrix} 0.1 \\ -3 \\ \vdots \\ 2.4 \\ 0.9 \end{bmatrix}$$


Each dimension of input vector represents some characteristics.

$$\begin{bmatrix} 3 \\ -3 \\ \vdots \\ 2.4 \\ 0.9 \end{bmatrix}$$


Longer hair

$$\begin{bmatrix} 0.1 \\ 2.1 \\ \vdots \\ 2.4 \\ 0.9 \end{bmatrix}$$


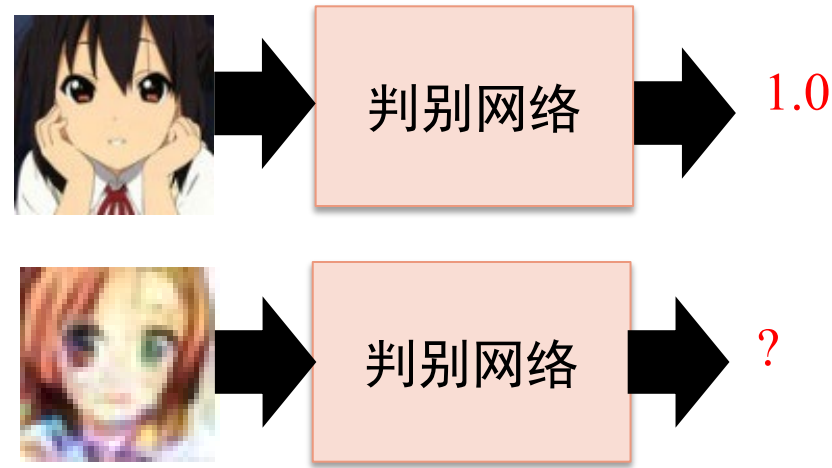
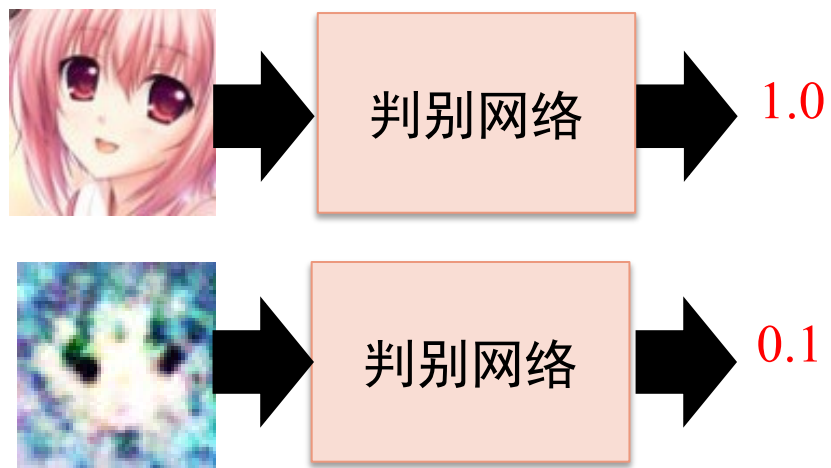
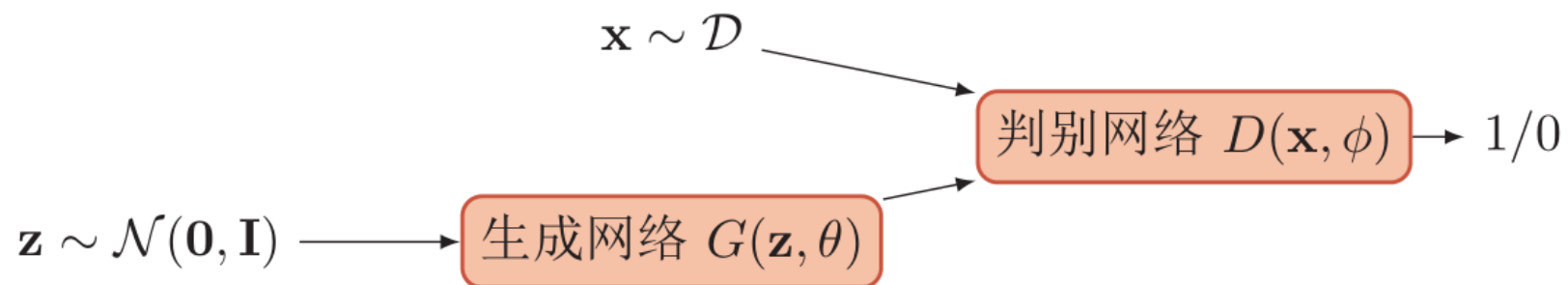
blue hair

$$\begin{bmatrix} 0.1 \\ -3 \\ \vdots \\ 2.4 \\ 3.5 \end{bmatrix}$$

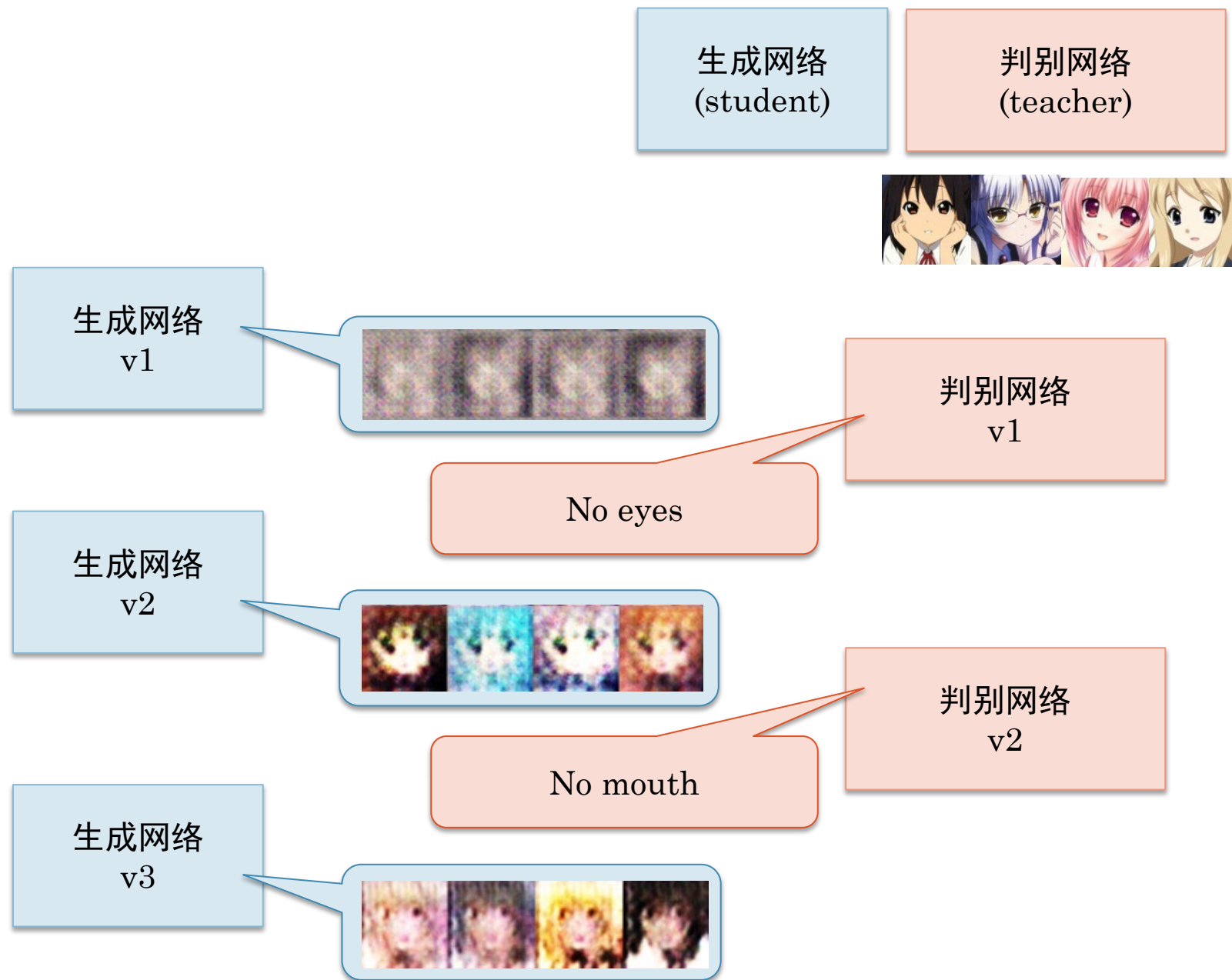

Open mouth

判别网络

- 判别网络的输入则为真实样本或生成网络的输出，其目的是将生成网络的输出从真实样本中尽可能分辨出来。



对抗过程



MinMax Game

• 判别网络

$$\max_{\phi} \mathbb{E}_{\mathbf{x} \sim p_r(\mathbf{x})} \left[\log D(\mathbf{x}; \phi) \right] + \mathbb{E}_{\mathbf{z} \sim p(\mathbf{z})} \left[\log(1 - D(G(\mathbf{z}; \theta); \phi)) \right]$$

将生成网络生成的样本与真实样本中尽可能区分出

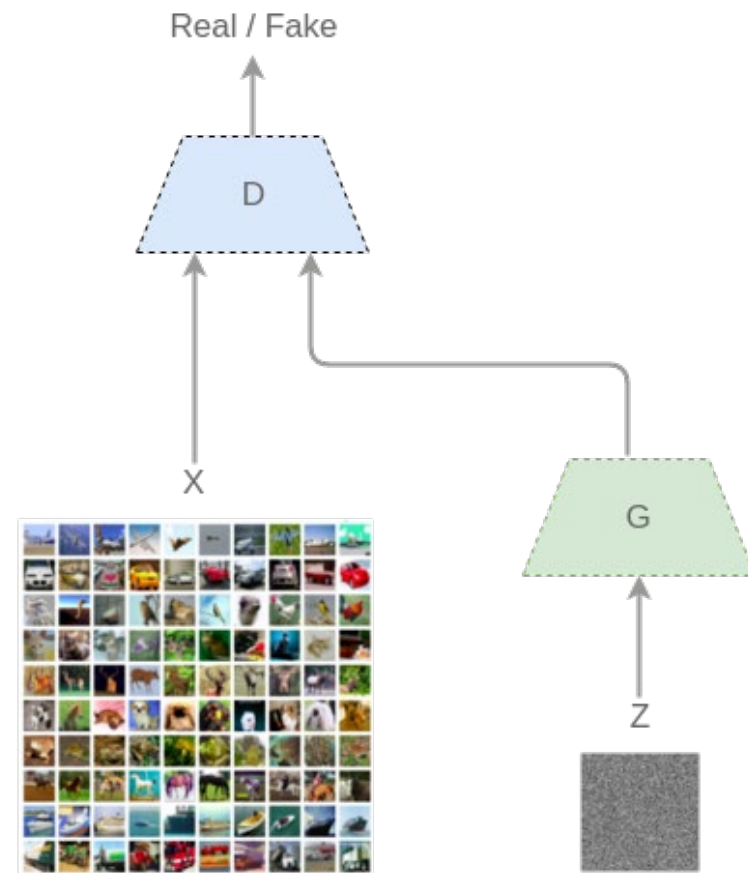
• 生成网络

$$\max_{\theta} \left(\mathbb{E}_{\mathbf{z} \sim p(\mathbf{z})} \left[\log D(G(\mathbf{z}; \theta); \phi) \right] \right)$$

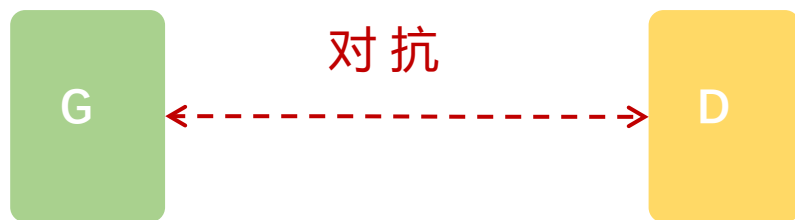
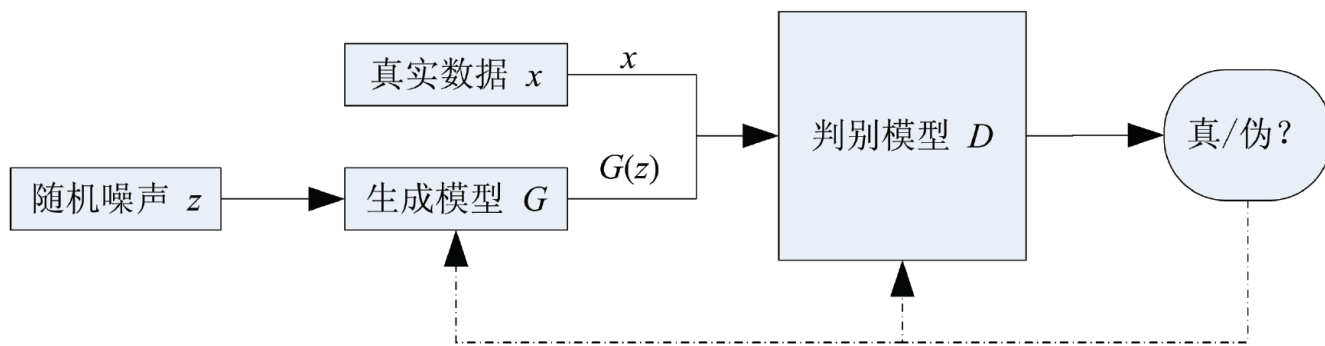
尽可能地欺骗判别网络

• Minmax Game

$$\min_{\theta} \max_{\phi} \left(\mathbb{E}_{\mathbf{x} \sim p_r(\mathbf{x})} \left[\log D(\mathbf{x}; \phi) \right] + \mathbb{E}_{\mathbf{z} \sim p(\mathbf{z})} \left[\log(1 - D(G(\mathbf{z}; \theta); \phi)) \right] \right)$$



生成对抗网络 (GAN)



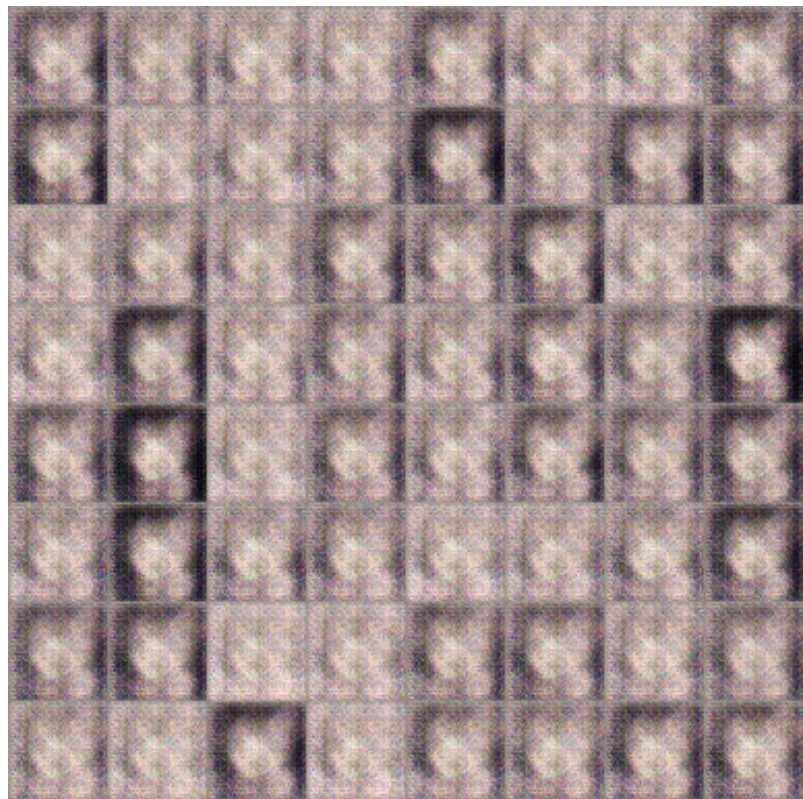
算法 13.1: 生成对抗网络的训练过程

输入: 训练集 $\mathcal{D}$, 对抗训练迭代次数 T , 每次判别网络的训练迭代次数 K , 小批量样本数量 M

```
1 随机初始化  $\theta, \phi$ ;  
2 for  $t \leftarrow 1$  to  $T$  do  
    // 训练判别网络  $D(\mathbf{x}, \phi)$   
3    for  $k \leftarrow 1$  to  $K$  do  
        // 采集小批量训练样本  
4        从训练集  $\mathcal{D}$  中采集  $M$  个样本  $\{\mathbf{x}^{(m)}\}, 1 \leq m \leq M$ ;  
5        从分布  $\mathcal{N}(\mathbf{0}, \mathbf{I})$  中采集  $M$  个样本  $\{\mathbf{z}^{(m)}\}, 1 \leq m \leq M$ ;  
6        使用随机梯度上升更新  $\phi$ , 梯度为  
            
$$\frac{\partial}{\partial \phi} \left[ \frac{1}{M} \sum_{m=1}^M \left( \log D(\mathbf{x}^{(m)}, \phi) + \log (1 - D(G(\mathbf{z}^{(m)}, \theta), \phi)) \right) \right];$$
  
7    end  
    // 训练生成网络  $G(\mathbf{z}, \theta)$   
8    从分布  $\mathcal{N}(\mathbf{0}, \mathbf{I})$  中采集  $M$  个样本  $\{\mathbf{z}^{(m)}\}, 1 \leq m \leq M$ ;  
9    使用随机梯度上升更新  $\theta$ , 梯度为  
            
$$\frac{\partial}{\partial \theta} \left[ \frac{1}{M} \sum_{m=1}^M D(G(\mathbf{z}^{(m)}, \theta), \phi) \right];$$
  
10 end
```

输出: 生成网络 $G(\mathbf{z}, \theta)$

Anime Face Generation



100 updates

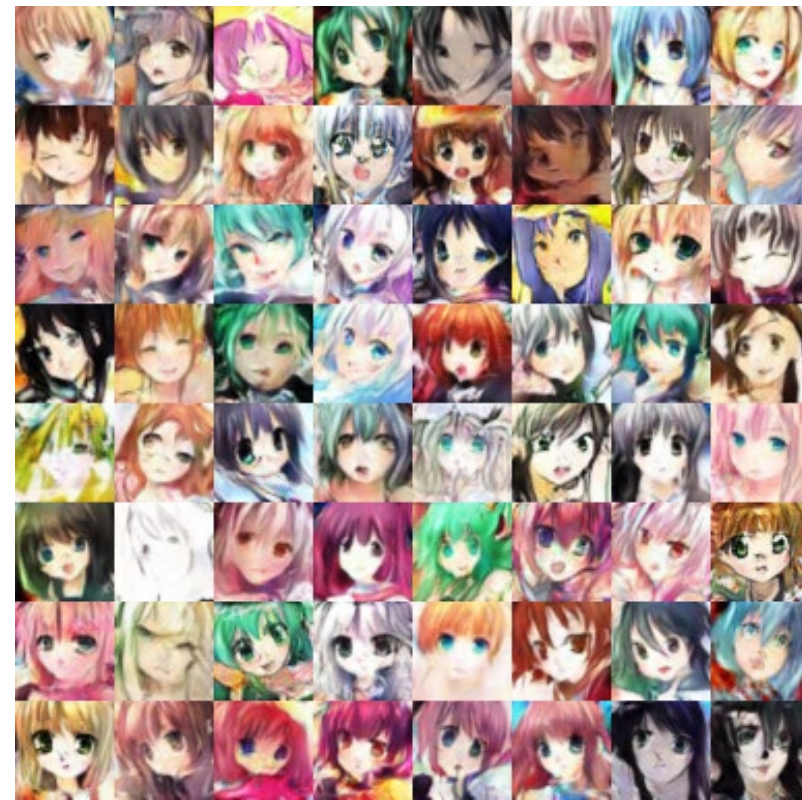


1000 updates

Anime Face Generation



10,000 updates

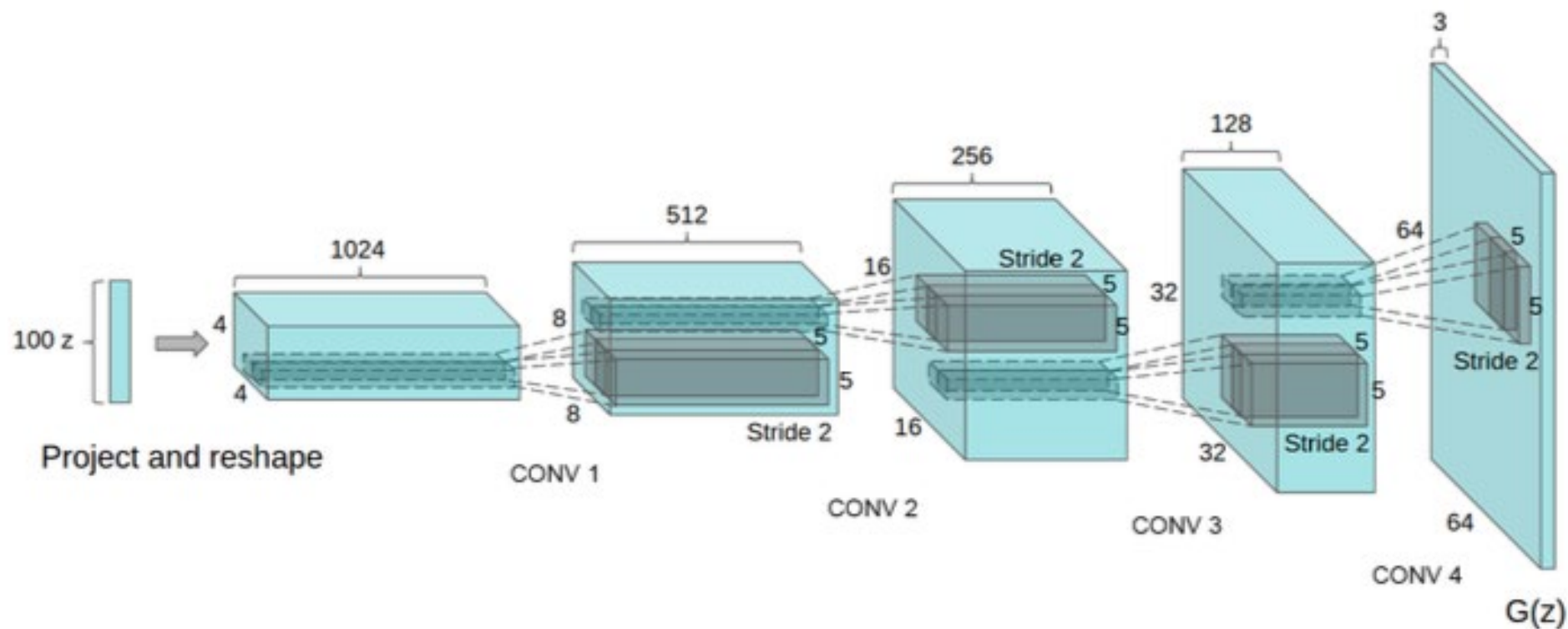


50,000 updates

无监督式生成对抗网络

- **Deep Convolution GAN (DCGAN)**

- 全部都是卷积层
- 需要使用 Virtual Batch Normalization
- LeakyReLU 作为激活函数



无监督式生成对抗网络

• DCGAN



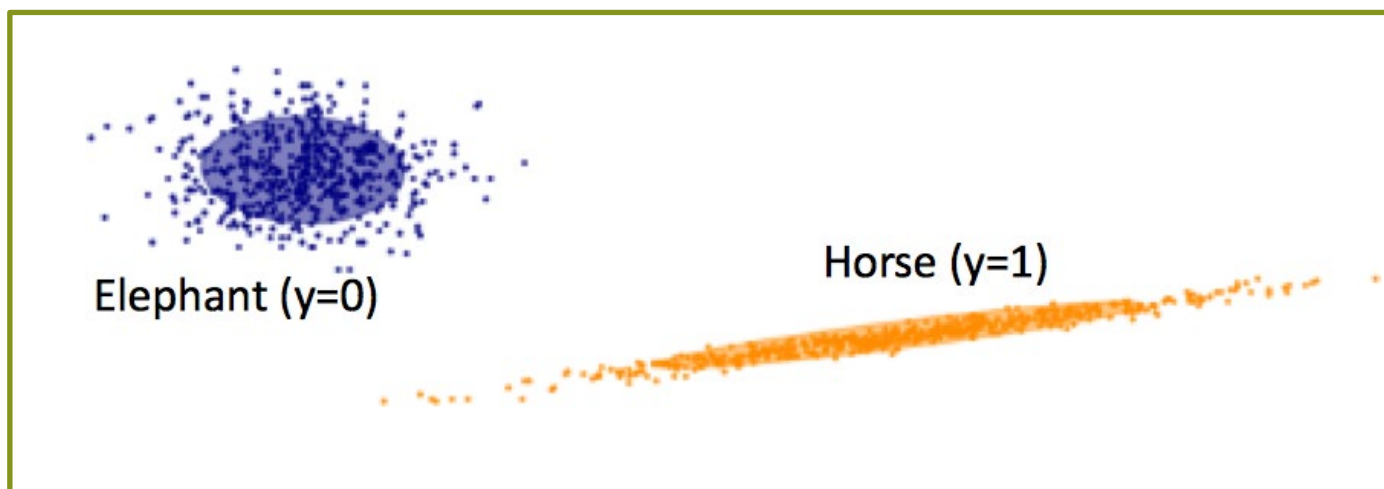
DCGANs for LSUN Bedrooms

无监督式生成对抗网络

• Density Estimation

- Discriminative model: $p(y|x)$
 - $y = 0$ (大象 elephant) , $y = 1$ (马, horse)
- Generative model: $p(x|y)$

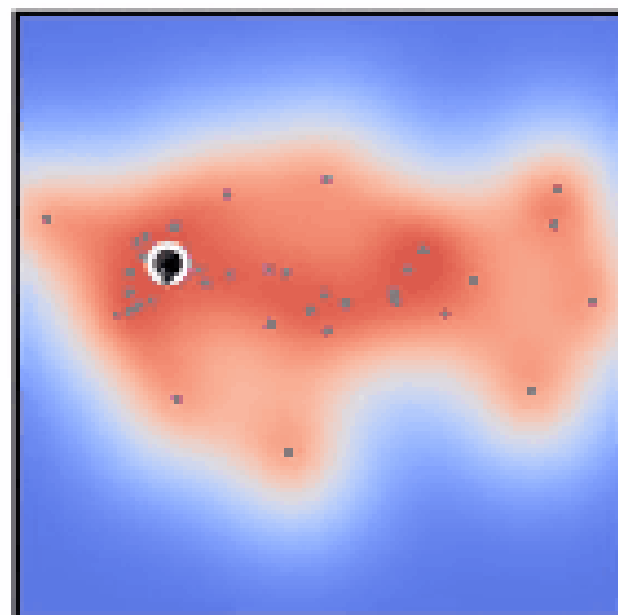
$$\mathbb{E}_{\mathbf{x} \sim p_r(\mathbf{x})}$$



curvature, thickness and angle, and the aspect ratio.

Interactive 2D Font Manifold Demonstration

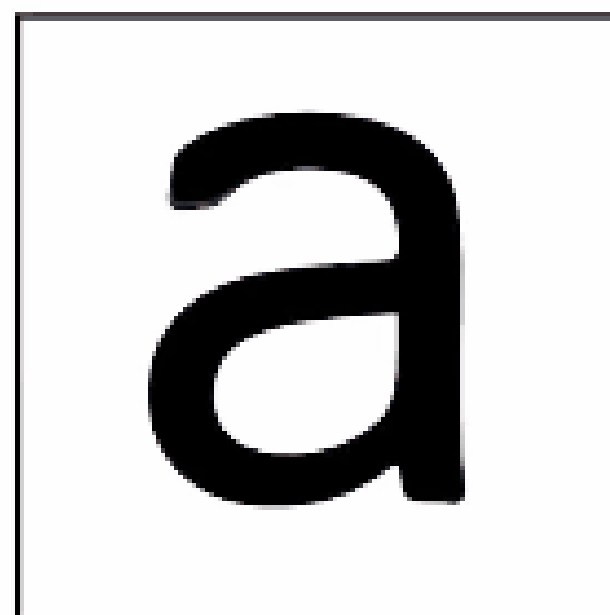
Please drag the black and white circle around the heat map to explore the 2D font manifold.



Unlikely

Probability

Likely



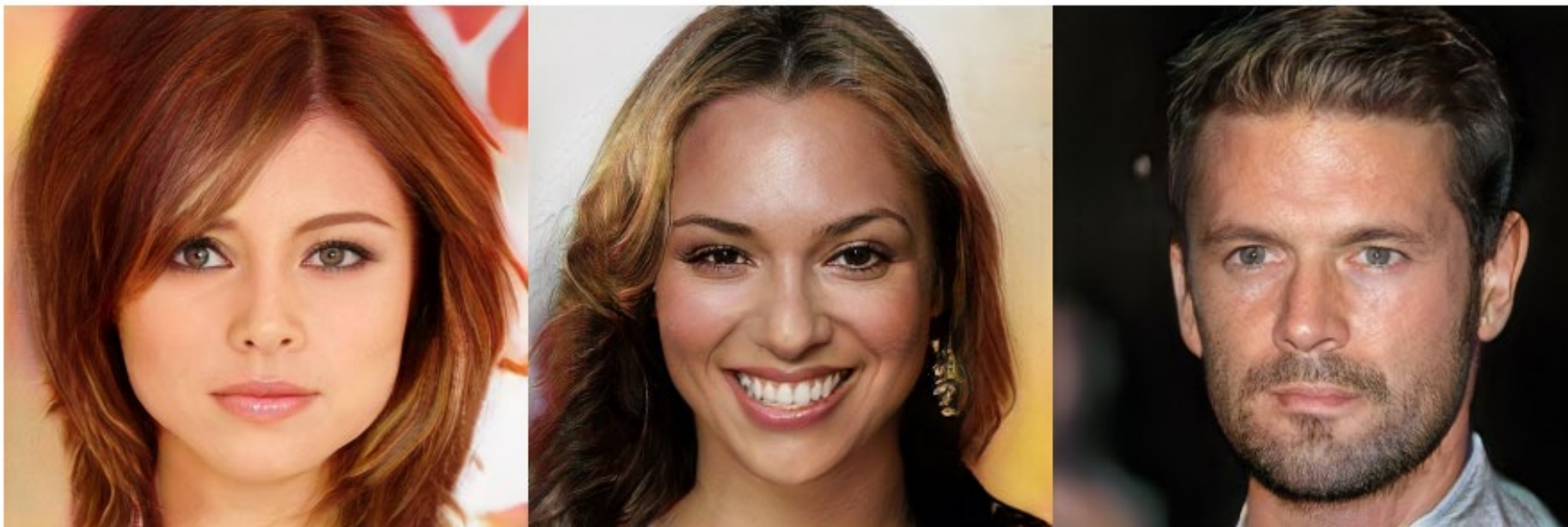
Select Character:

a f

Source: [Campbell et al 2014], From an earlier paper

无监督式生成对抗网络

- 示例：生成人脸 (PG-GAN, BigGAN, StyleGAN etc.)



无监督式生成对抗网络



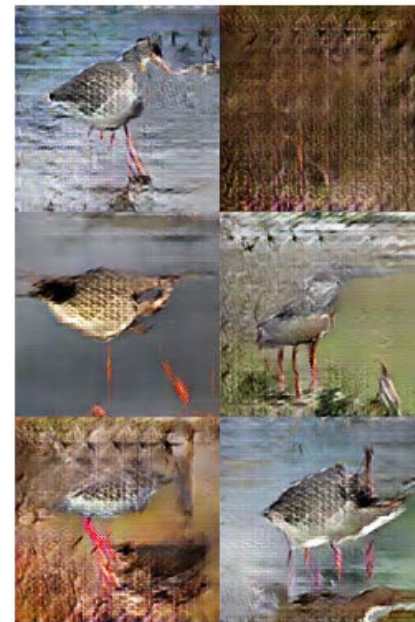
monarch butterfly



goldfinch



daisy

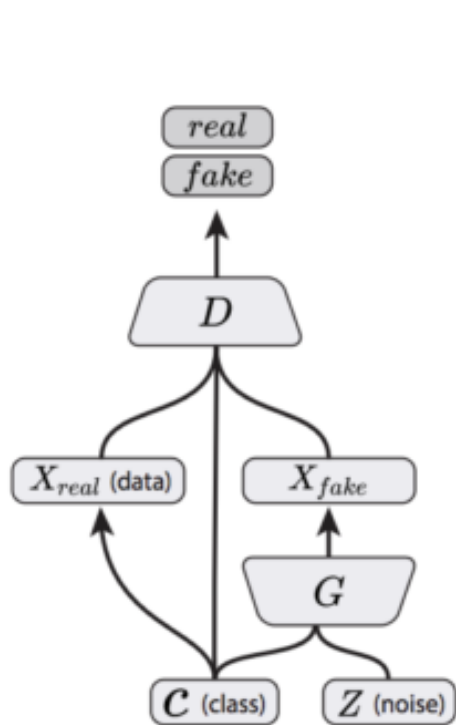


redshank

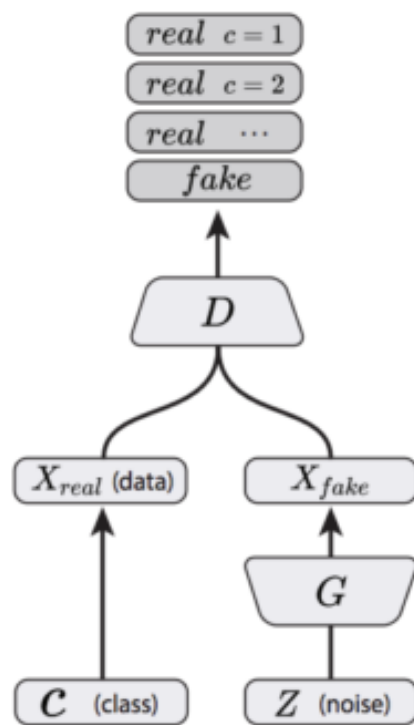
无监督式生成对抗网络

• AC-GAN

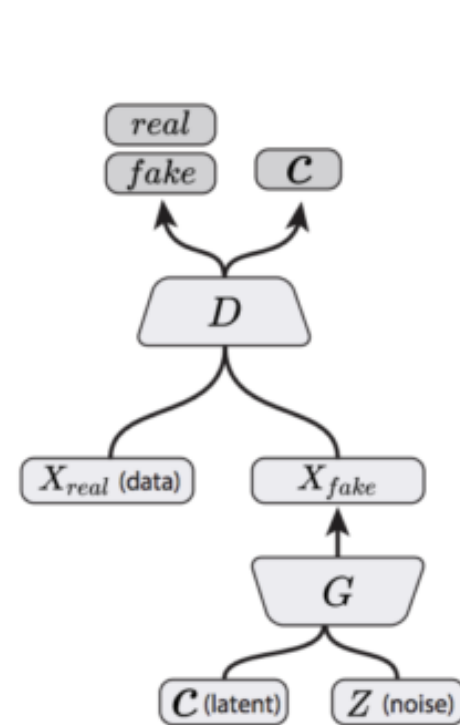
- Conditional Image Synthesis with Auxiliary Classifier GANs



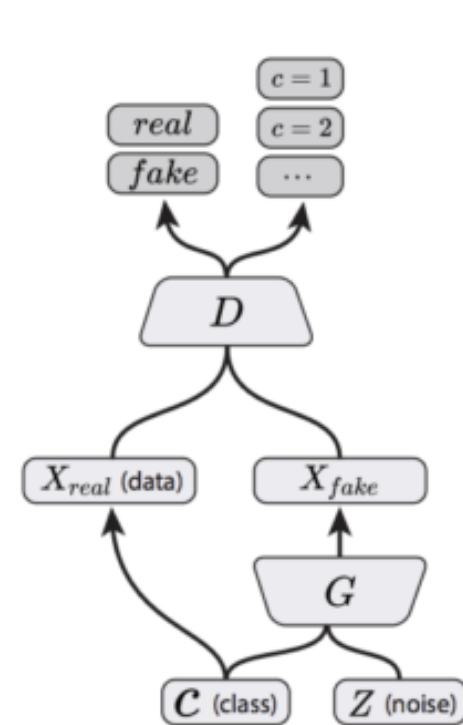
Conditional GAN
(Mirza & Osindero, 2014)



Semi-Supervised GAN
(Odena, 2016; Salimans, et al., 2016)



InfoGAN
(Chen, et al., 2016)



AC-GAN
(Present Work)

Mode Collapse



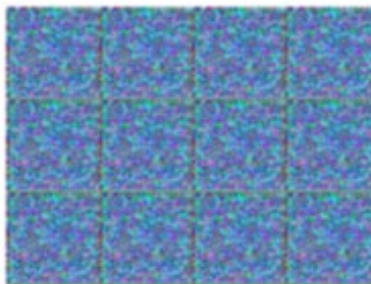
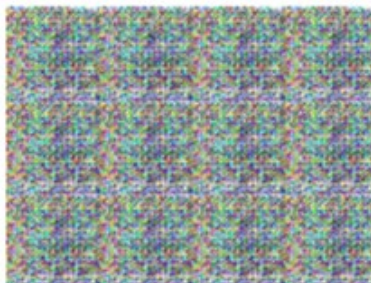
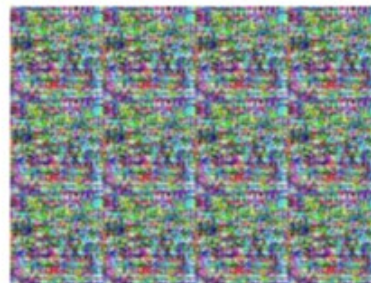
无监督式生成 对抗网络

DCGAN

G: MLP, D: CNN



LSGAN



Original
WGAN



Improved
WGAN



Wasserstein GAN

Wasserstein距离

- Wasserstein距离用于衡量两个分布之间的距离。

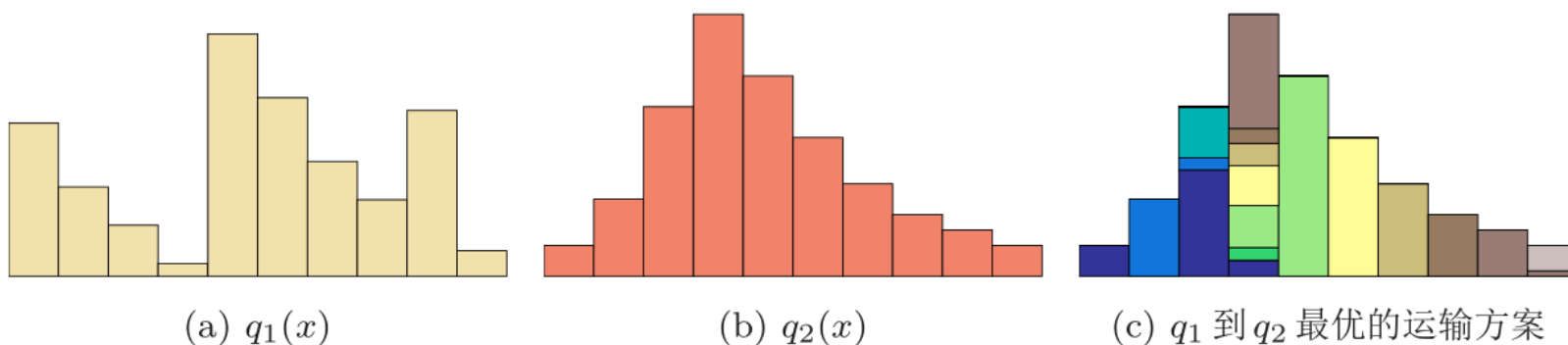
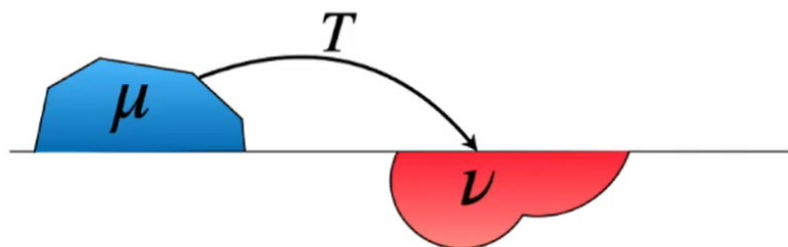
$$W_p(q_1, q_2) = \left(\inf_{\gamma(x,y) \in \Gamma(q_1, q_2)} \mathbb{E}_{(x,y) \sim \gamma(x,y)} [d(x,y)^p] \right)^{\frac{1}{p}}$$

- 其中 $\Gamma(q_1, q_2)$ 是边际分布为 q_1, q_2 的所有可能的联合分布集合, $d(x,y)$ 为 x 和 y 的距离, 比如 ℓ_p 距离等。
- Wasserstein距离相比KL散度和JS散度的优势在于: 即使两个分布没有重叠或者重叠非常少, Wasserstein距离仍然能反映两个分布的远近。

Wasserstein距离

• Wasserstein距离

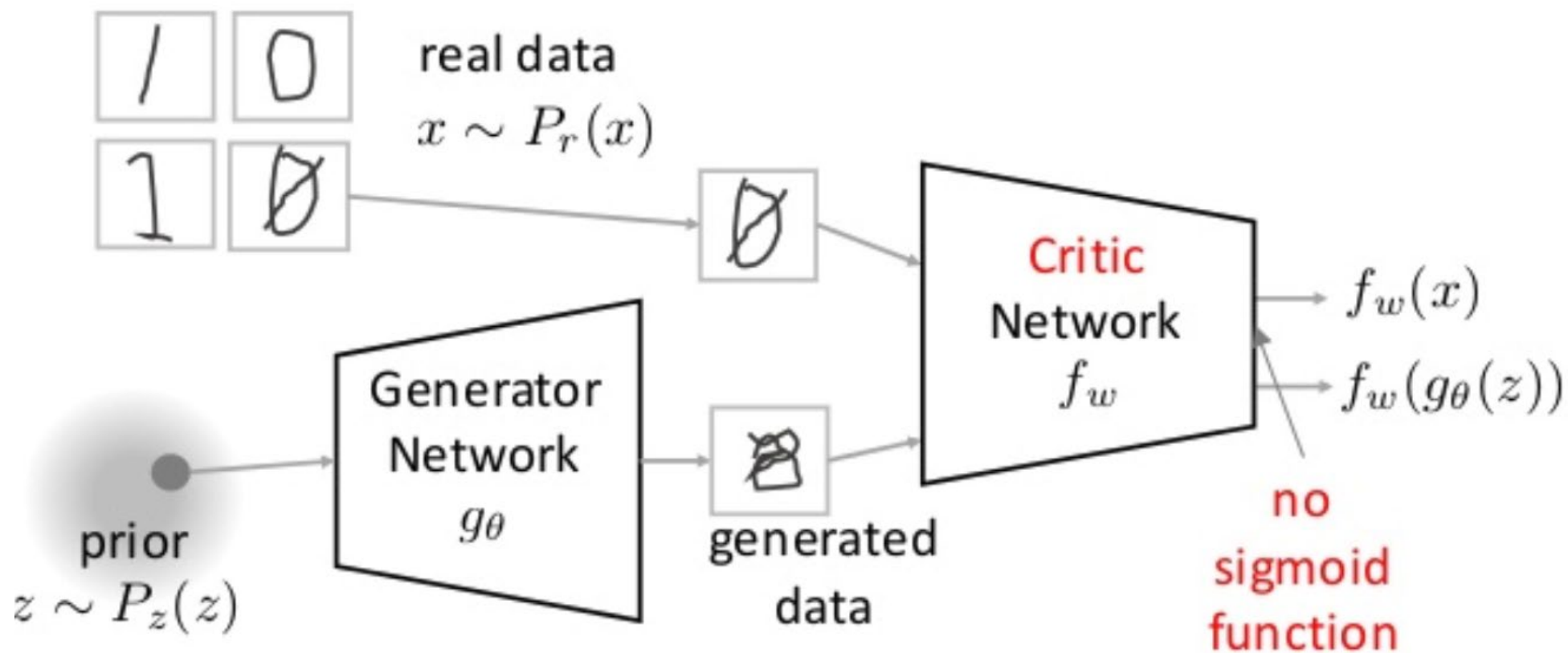
- 如果将两个分布看作是两个土堆，联合分布 $\gamma(x,y)$ 看作是从土堆 q_1 的位置 x 到土堆 q_2 的位置 y 的搬运土的数量。Wasserstein距离可以理解为搬运土堆的最小工作量，也称为推土机距离（Earth-Mover's Distance, EMD）。



Wasserstein GAN

$$\min_{\theta} \max_{w \in [-k, k]^l} \mathbb{E}_{x \sim P_r} [f_w(x)] - \mathbb{E}_{z \sim P_z} [f_w(g_{\theta}(z))]$$

k-Lipschitz Constraint



Wasserstein GAN

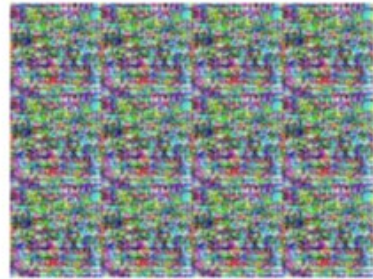
DCGAN

LSGAN

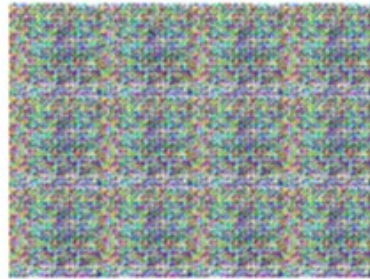
Original
WGAN

Improved
WGAN

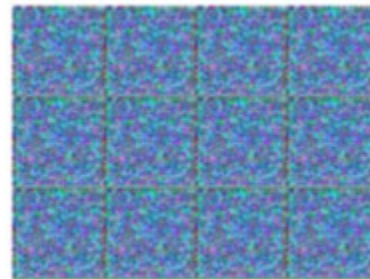
G: MLP, D: CNN



G: CNN (bad structure), D: CNN



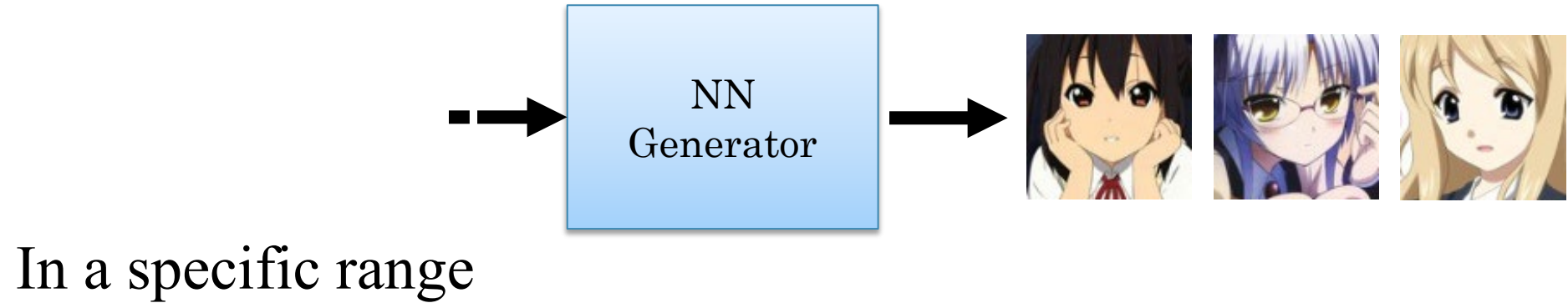
G: 101 layer, D: 101 layer



条件式GAN

Conditional Generation

Generation



Conditional Generation

“Girl with red hair and red eyes”

“Girl with yellow ribbon”



Conditional Generation

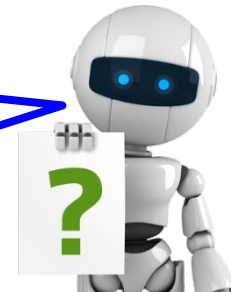
- We don't want to simply generate some random stuff.
- Generate objects based on conditions:

Caption Generation

Given
condition:

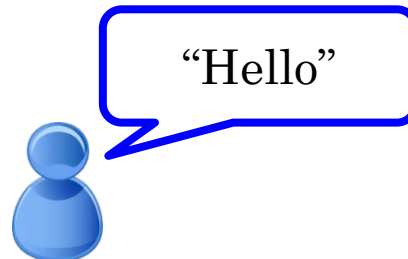


“A young girl is
dancing.”

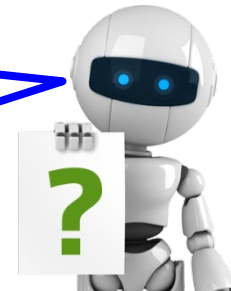


Chat-bot

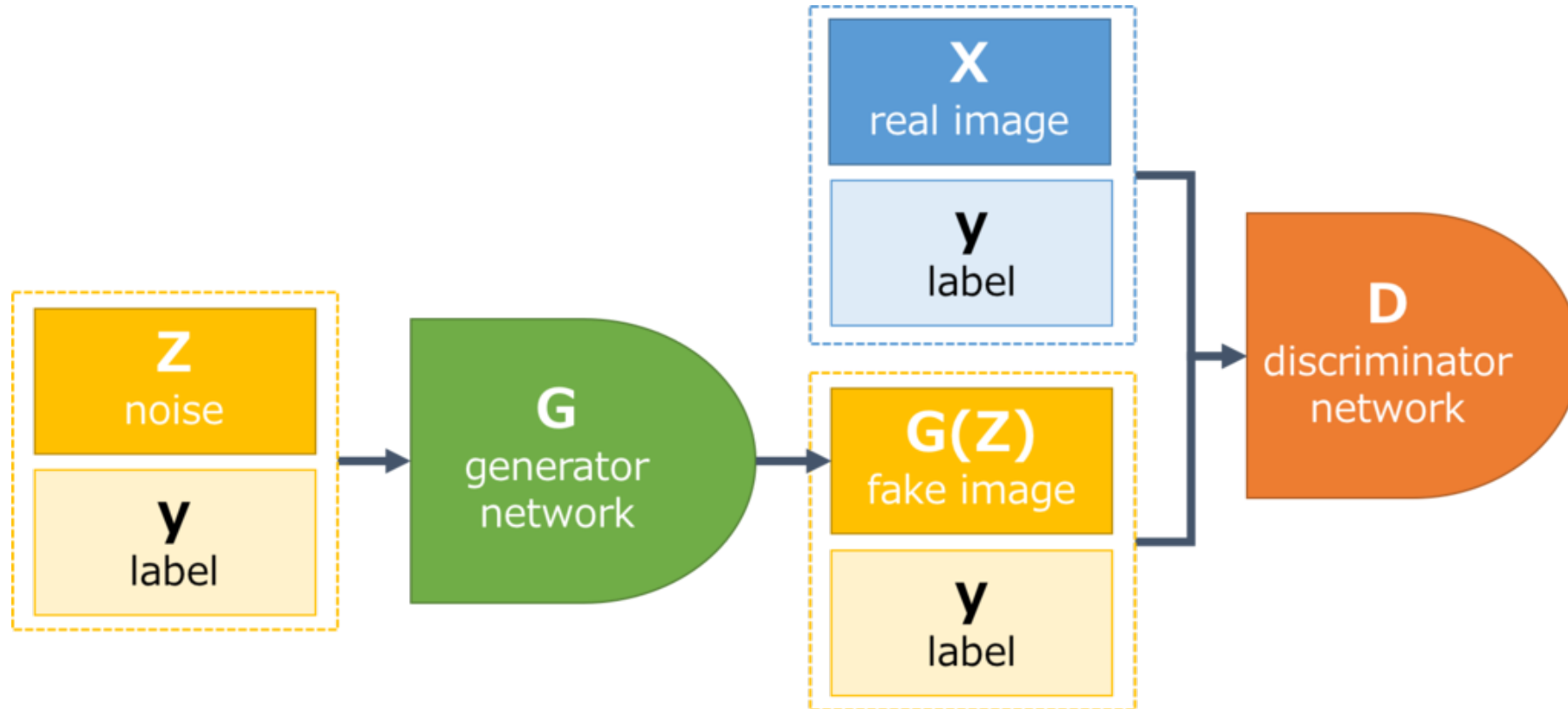
Given
condition:



“Hello. Nice to
see you.”

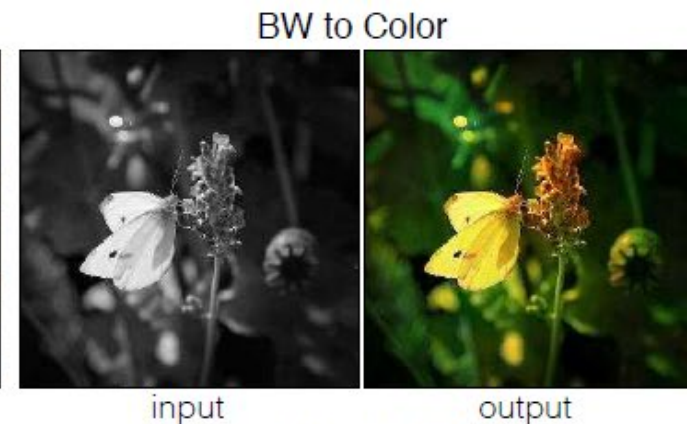
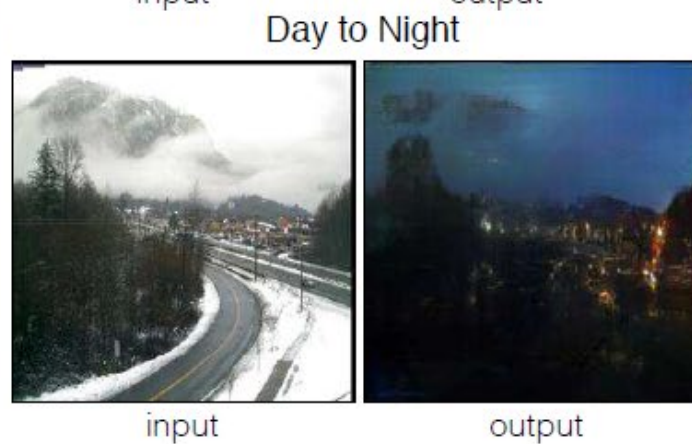
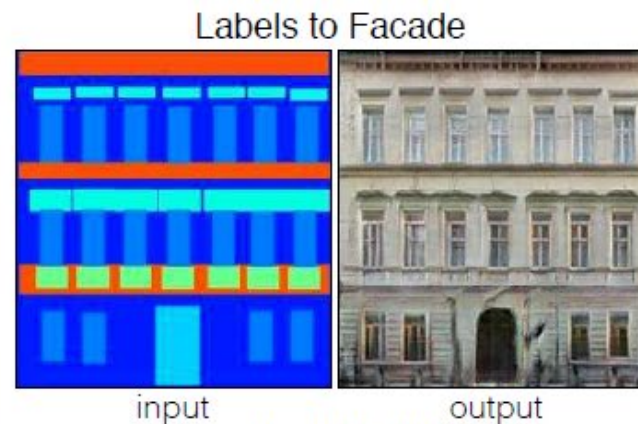
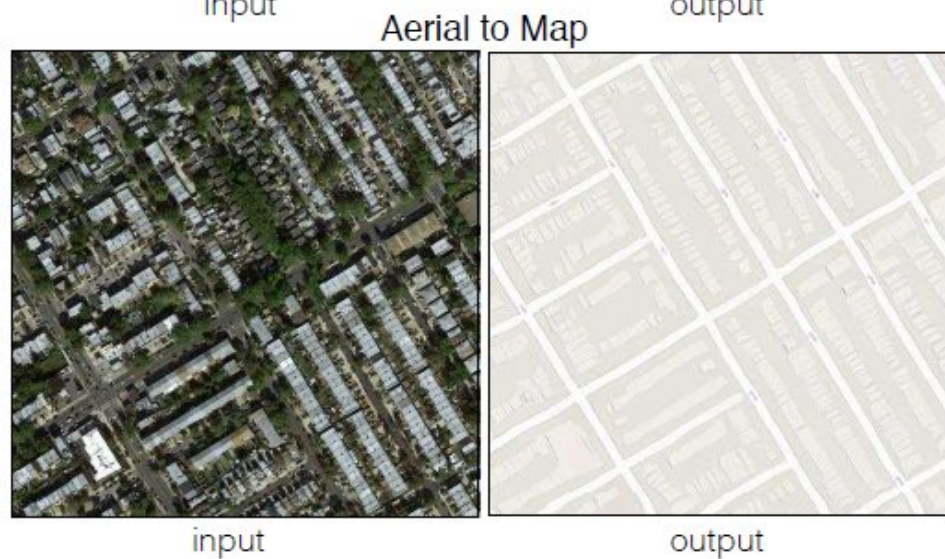


Conditional GAN



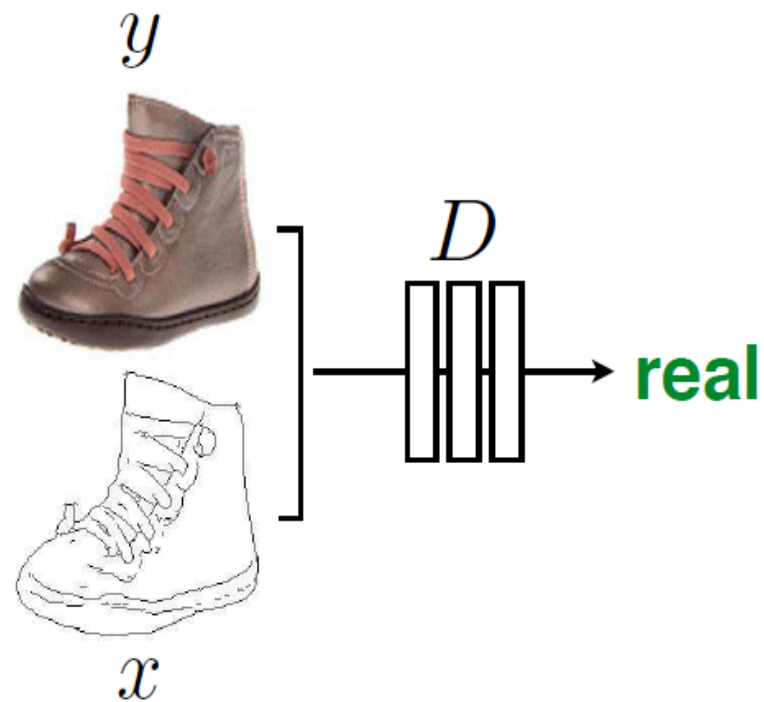
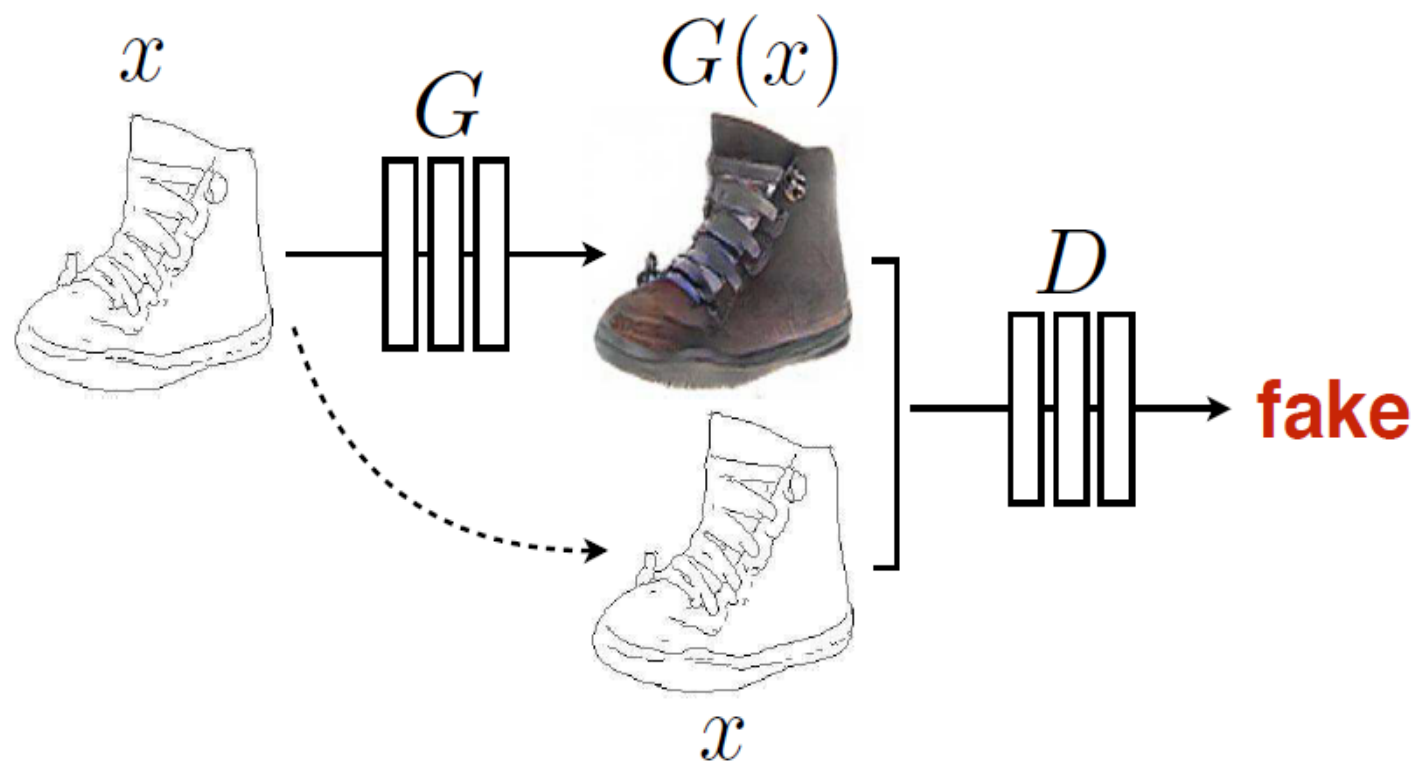
条件式生成对抗网络 Conditional GAN

• Image-to-Image Translation



条件式生成对抗网络 Conditional GAN

• Pix2Pix



条件式生成对抗网络 Conditional GAN

- U-Net (Pix2Pix)

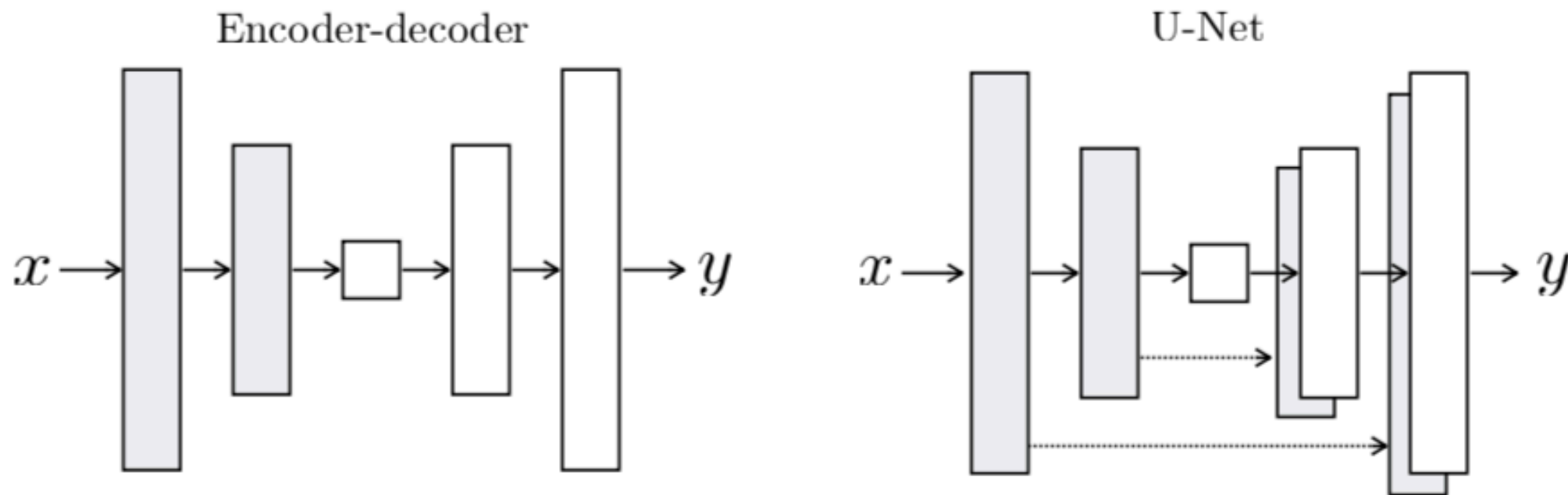
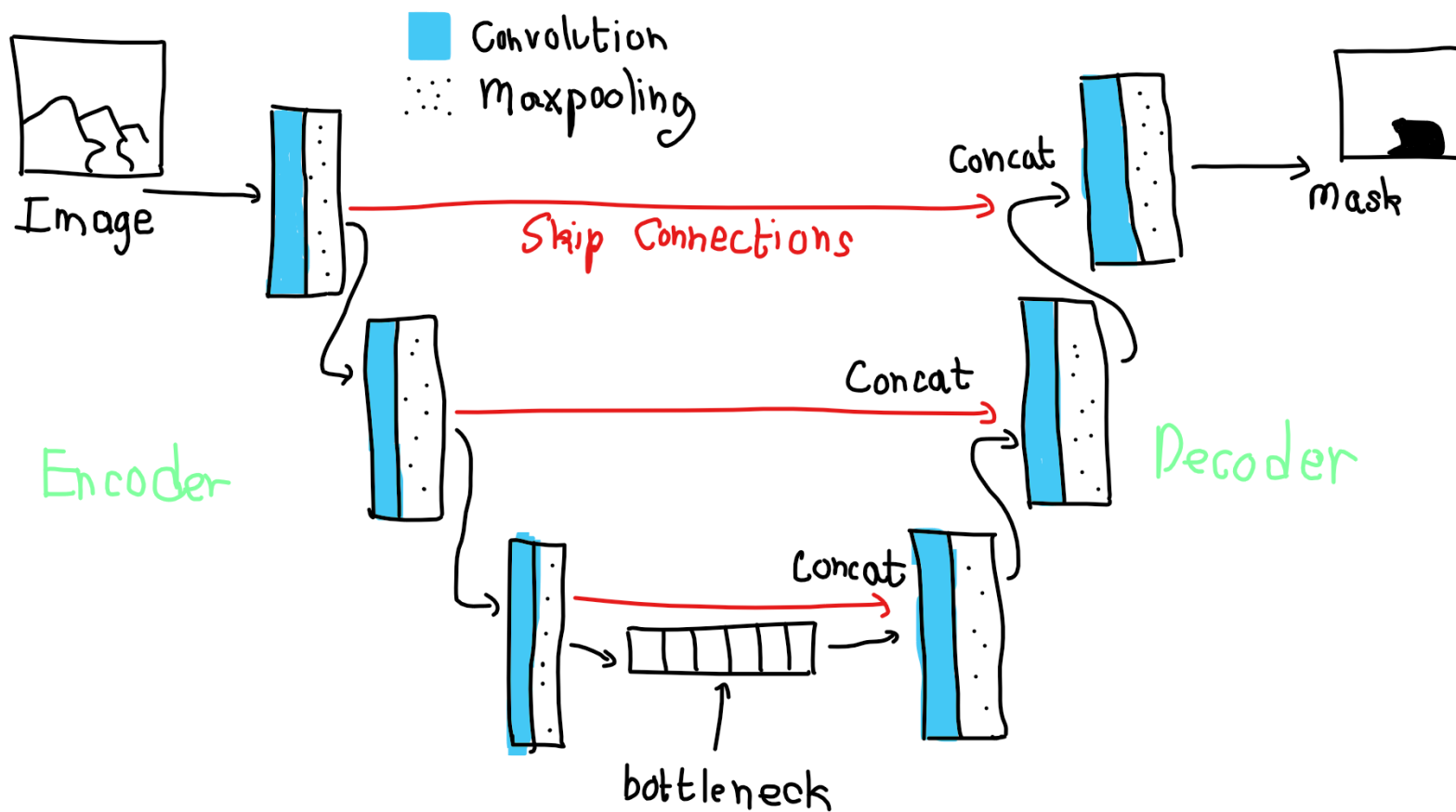


Figure 3: Two choices for the architecture of the generator. The “U-Net” [50] is an encoder-decoder with skip connections between mirrored layers in the encoder and decoder stacks.

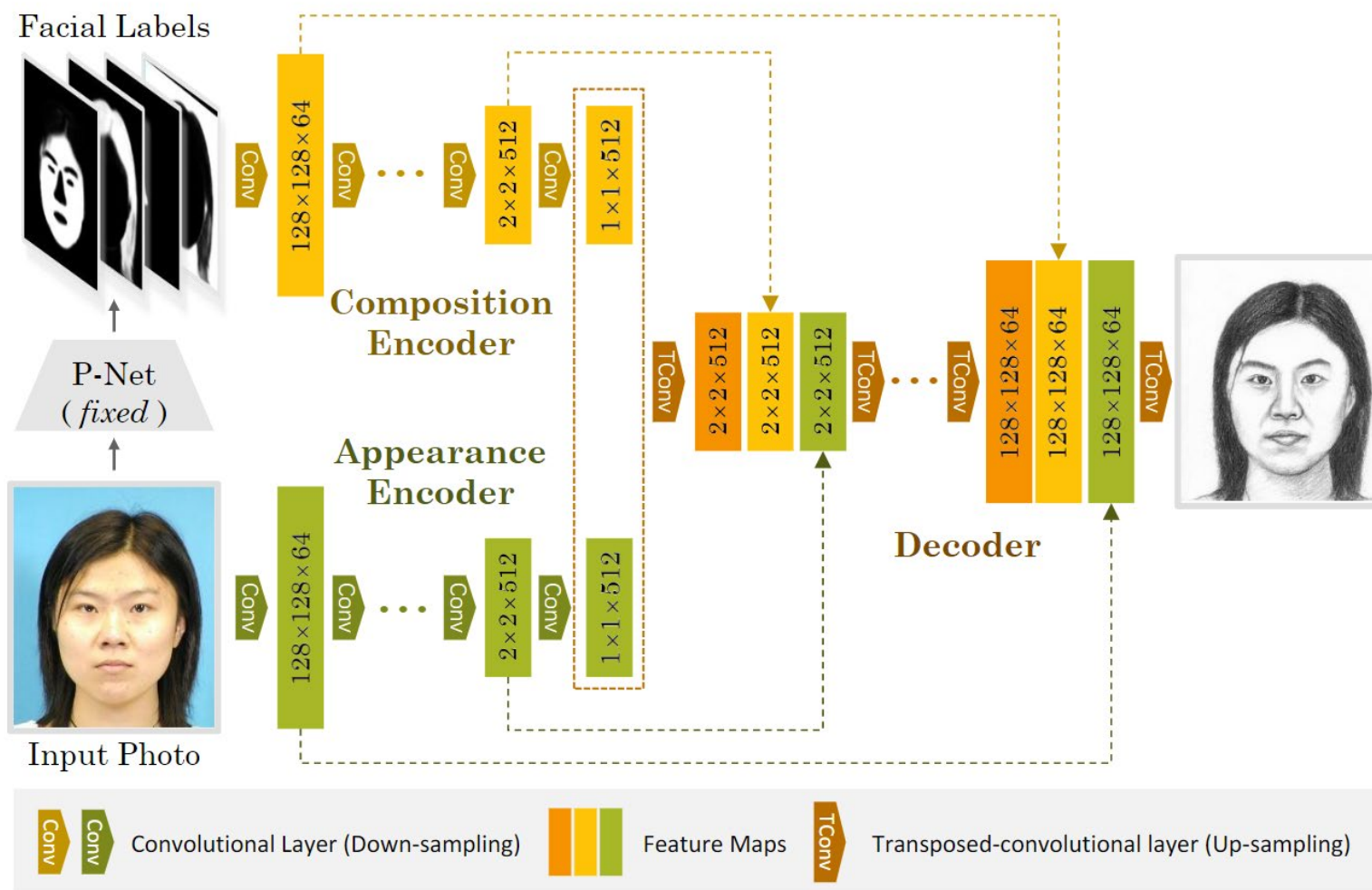
条件式生成对抗网络 Conditional GAN

• U-Net



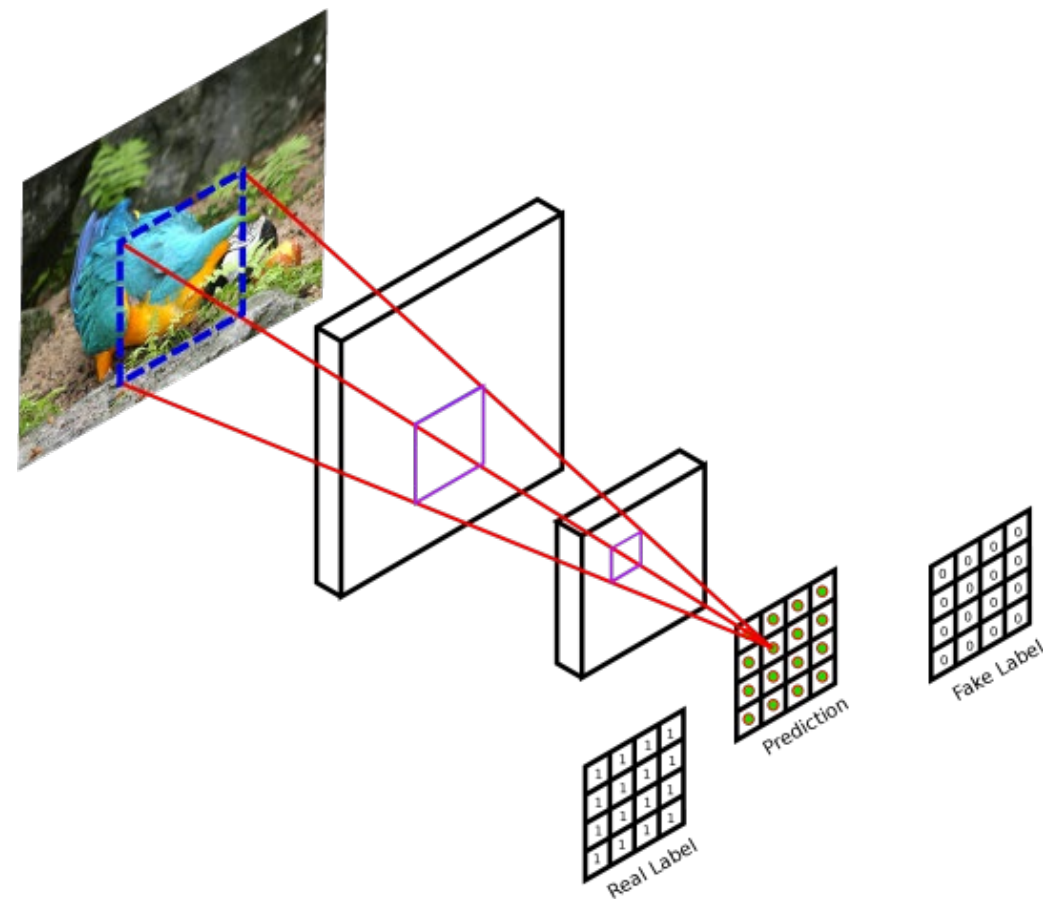
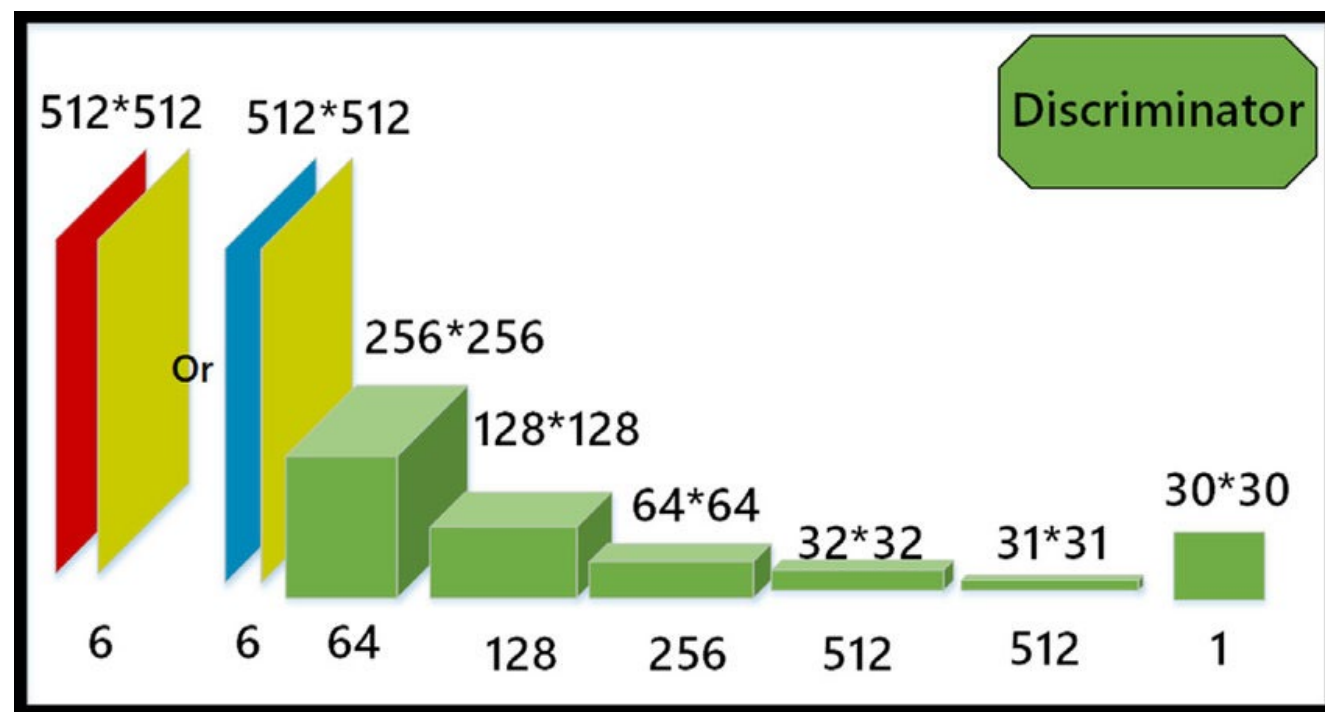
条件式生成对抗网络 Conditional GAN

• U-Net



条件式生成对抗网络 Conditionanl GAN

- Discriminator Structure: PatchGAN (Pix2Pix)



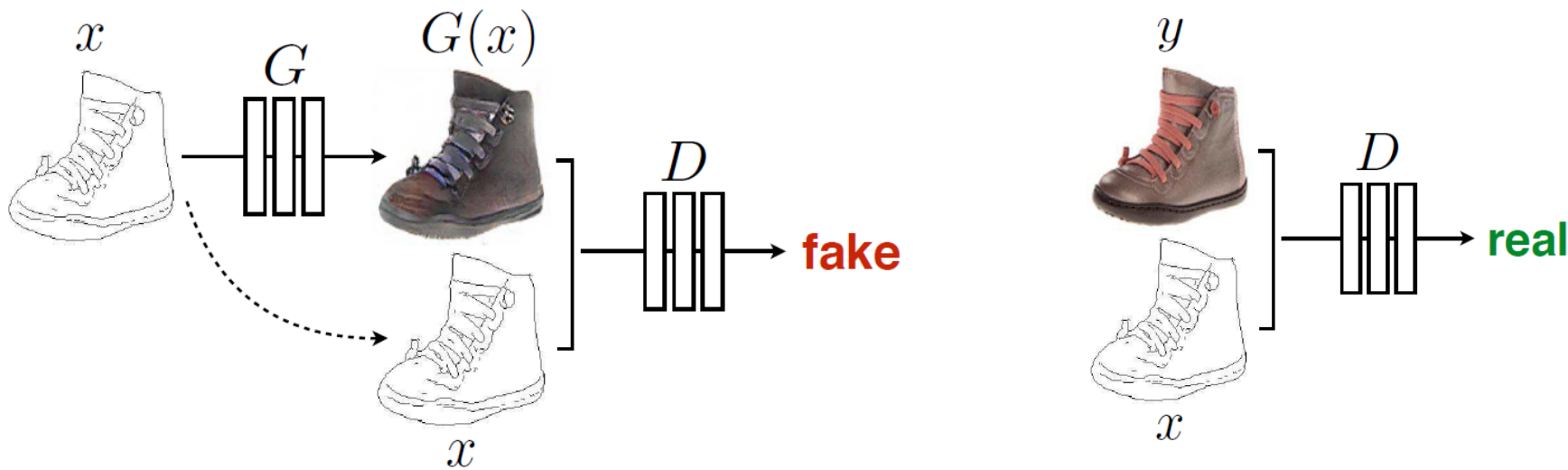
条件式生成对抗网络 Conditional GAN

- Loss Function: **Adversarial Loss + L1 Reconstruction Loss**

$$G^* = \arg \min_G \max_D \mathcal{L}_{cGAN}(G, D) + \lambda \mathcal{L}_{L1}(G).$$

$$\max_{\theta} \left(\mathbb{E}_{z \sim p(z)} \left[\log D(G(z; \theta); \phi) \right] \right)$$

$$\mathcal{L}_{L1}(G) = \mathbb{E}_{x, y, z} [\|y - G(x, z)\|_1].$$

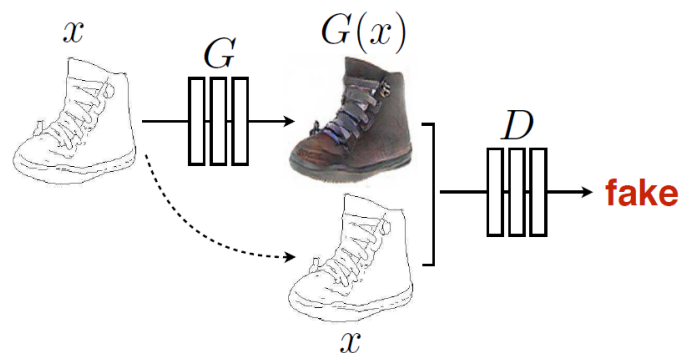


条件式生成对抗网络 Conditional GAN

$$G^* = \arg \min_G \max_D \mathcal{L}_{cGAN}(G, D) + \lambda \mathcal{L}_{L1}(G).$$

$$\mathcal{L}_{L1}(G) = \mathbb{E}_{x,y,z} [\|y - G(x, z)\|_1].$$

$$\max_{\theta} \left(\mathbb{E}_{z \sim p(z)} \left[\log D(G(z; \theta); \phi) \right] \right)$$



```
222 # netG
223 netG.zero_grad()
224 fake_AB = torch.cat((real_A, fake_B), 1)
225 out_put = netD(fake_AB)
226 loss_G_GAN = criterionGAN(out_put, True)
227
228 loss_G_L1 = critertion1(fake_B, real_B) * lamb #对于损失不同的权重
229 loss_G = loss_G_GAN + loss_G_L1
230 loss_G.backward()
231 optimizerG.step()
```


条件式生成对抗网络 Conditional GAN

• 实验结果

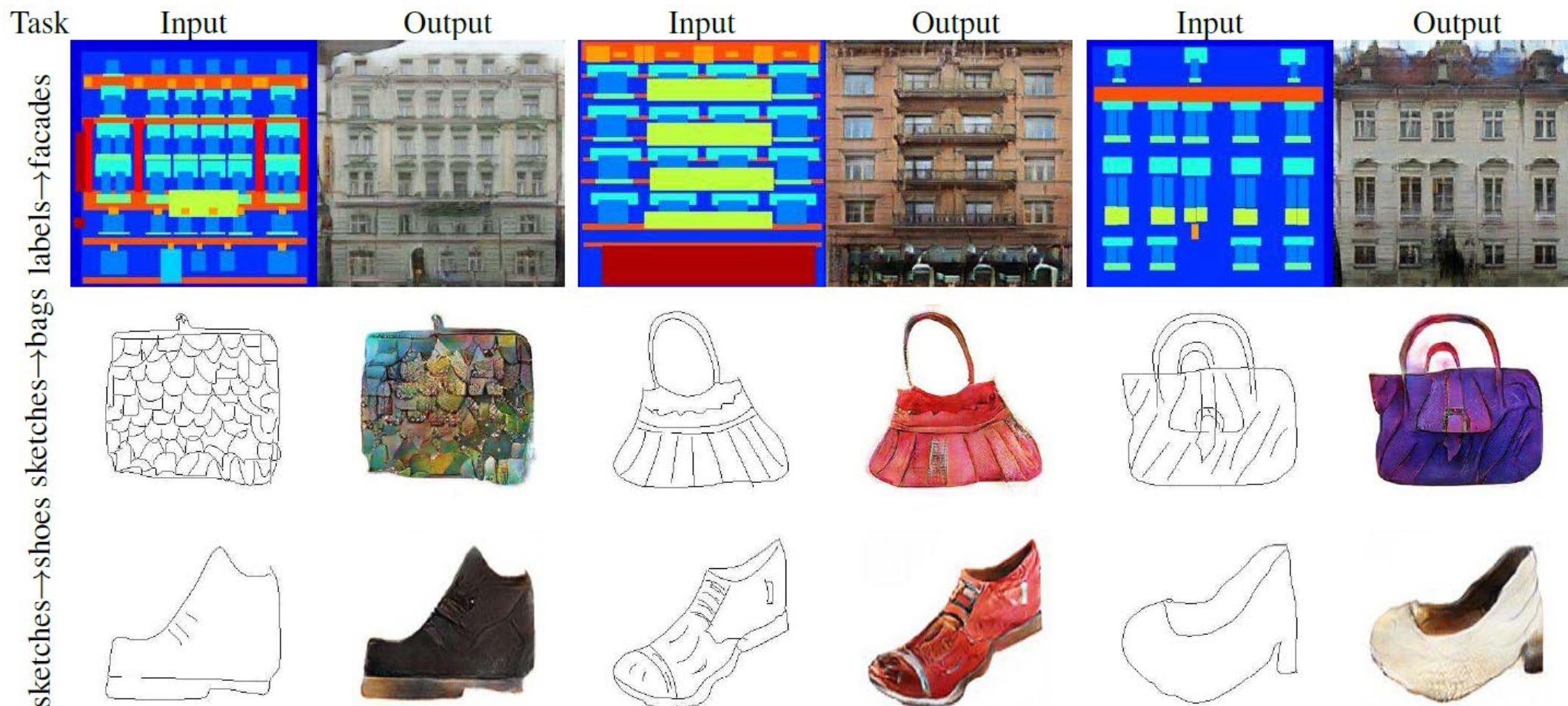
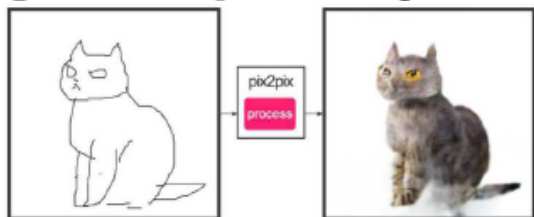


Figure 9: Results of our method on several tasks (data from [42] and [17]). Note that the sketch→photo results are generated by a model trained on automatic edge detections and tested on human-drawn sketches. Please see online materials for additional examples.

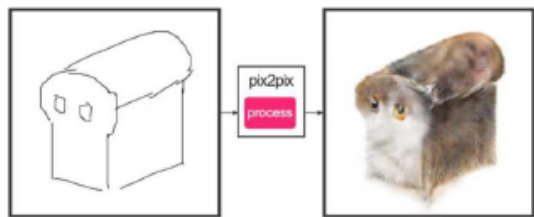
条件式生成对抗网络 Conditional GAN

• 实验结果

#edges2cats by Christopher Hesse



by @gods_tail



by @ivymyt



by @vvid

Sketch → Portrait



by Mario Klingemann

Depth → Streetview



by Jasper van Loenen

Background removal



by Kaihu Chen

“Do as I do”



by Brannon Dorsey

Palette generation



by Jack Qiao

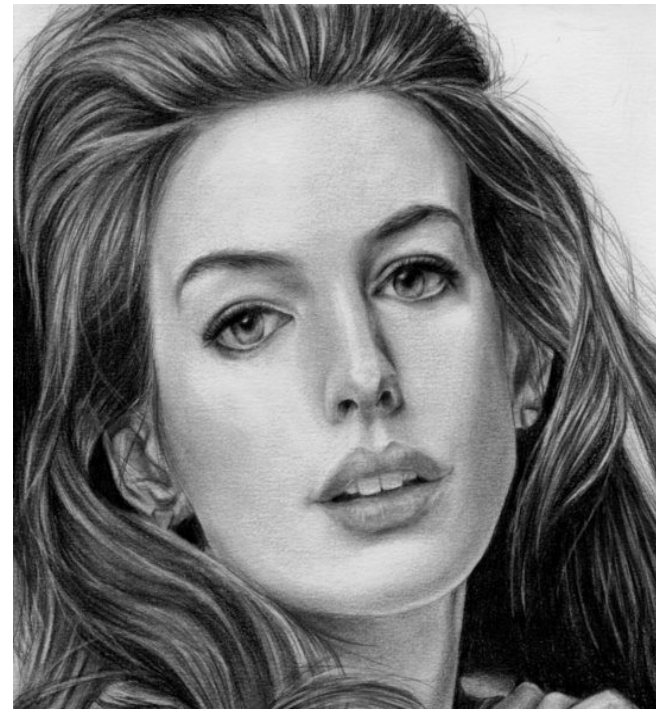
Sketch → Pokemon



by Bertrand Gondouin

Figure 10: Example applications developed by online community based on our pix2pix codebase: #edges2cats [3] by Christopher Hesse, Sketch → Portrait [7] by Mario Kingemann, “Do As I Do” pose transfer [2] by Brannon Dorsey, Depth → Streetview [5] by Jasper van Loenen, Background removal [6] by Kaihu Chen, Palette generation [4] by Jack Qiao, and Sketch → Pokemon [1] by Bertrand Gondouin.

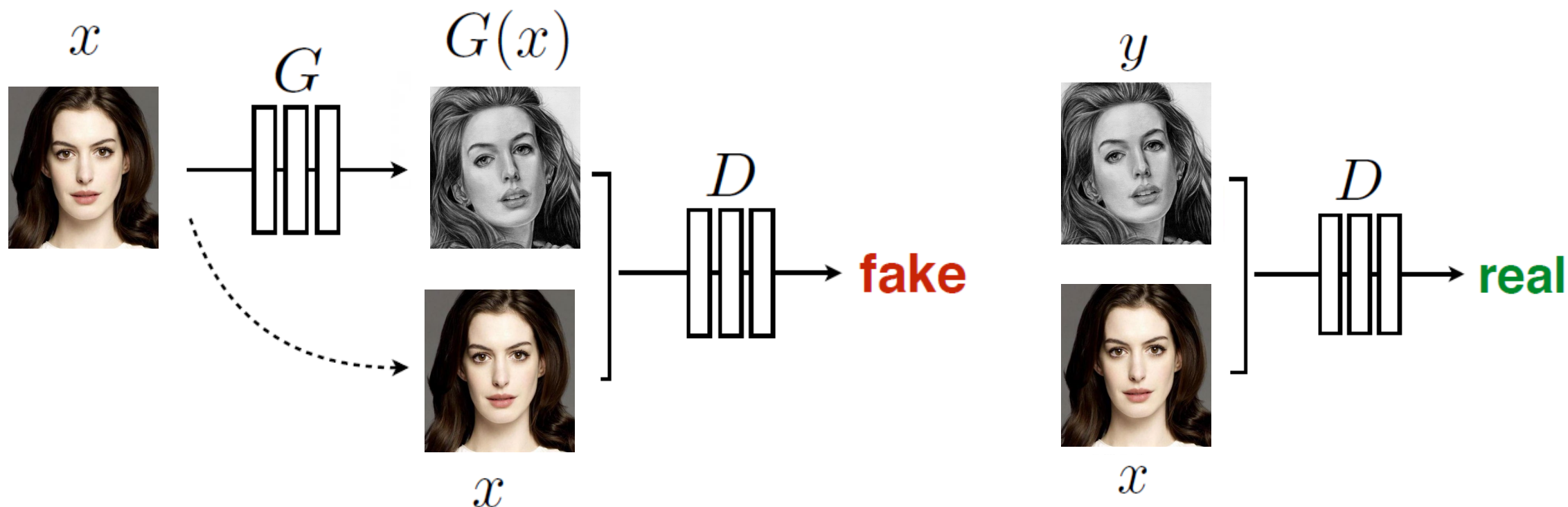
- Face Photo-Sketch Synthesis



异质人脸合成

• Face Photo-Sketch Synthesis

■ 框架 Framework



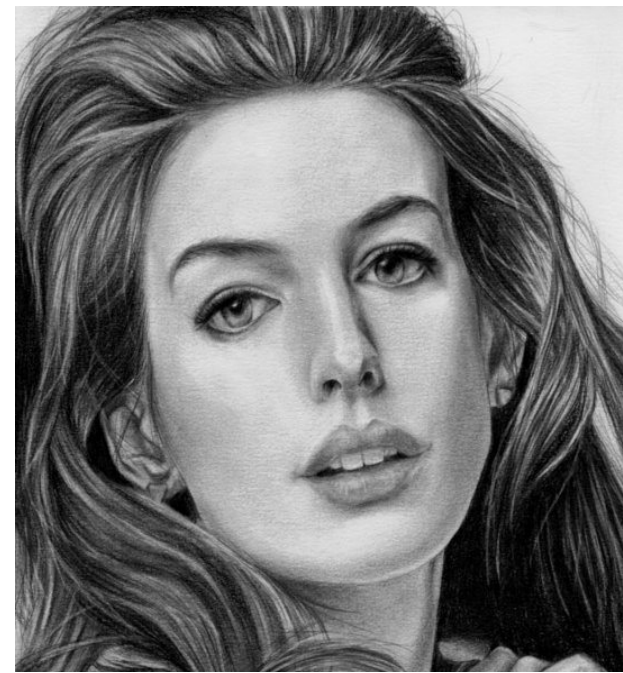
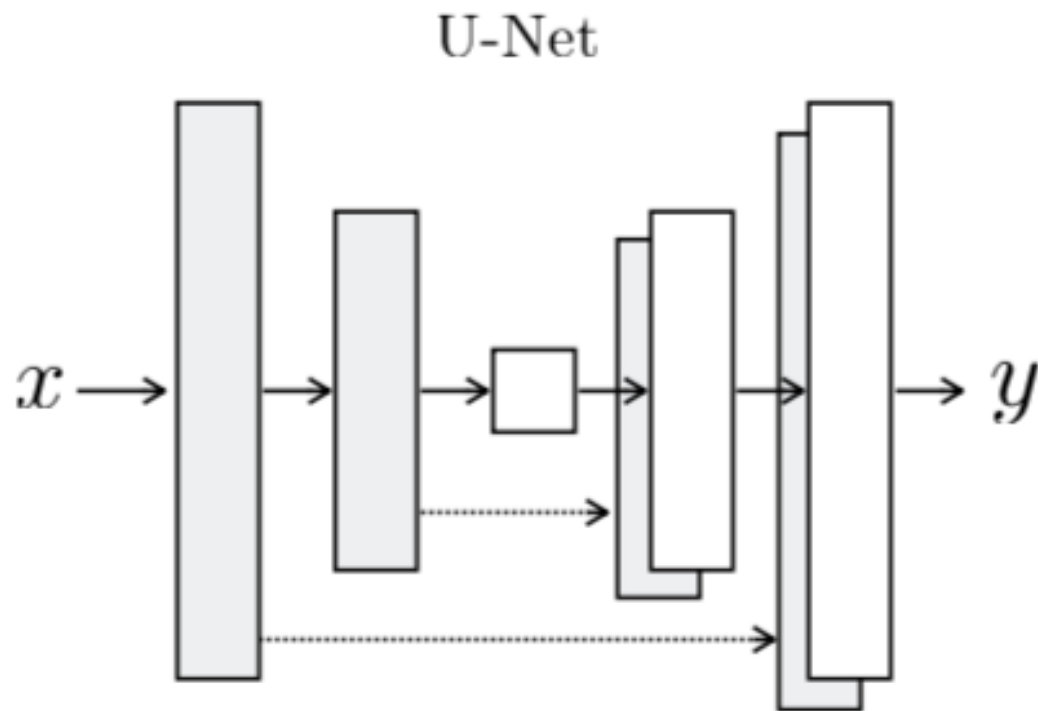
$$G^* = \arg \min_G \max_D \mathcal{L}_{cGAN}(G, D) + \lambda \mathcal{L}_{L1}(G).$$

$$\mathcal{L}_{L1}(G) = \mathbb{E}_{x,y,z} [\|y - G(x, z)\|_1].$$

异质人脸合成

• Face Photo-Sketch Synthesis

- 生成器 Generator

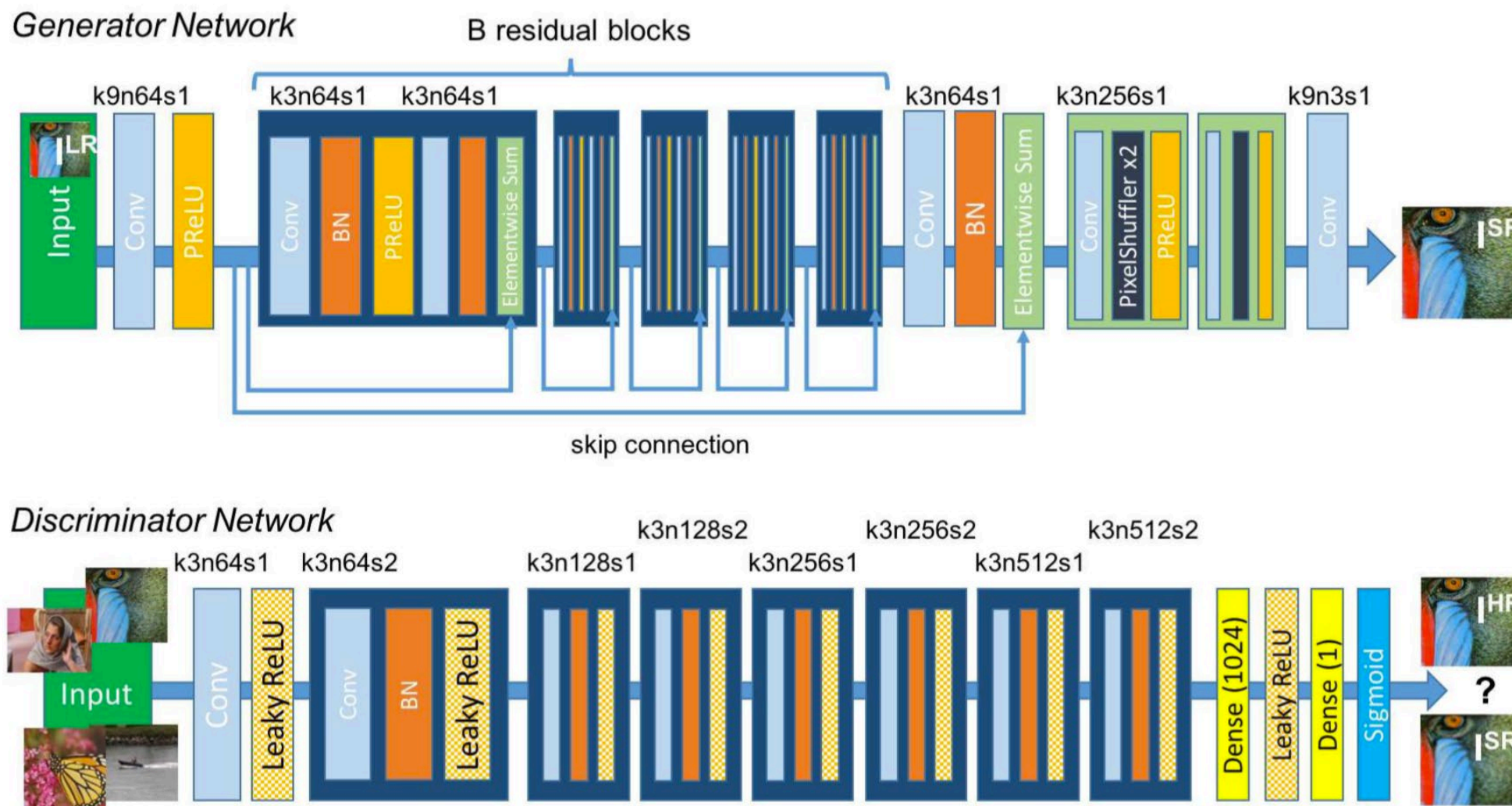


图像超分辨率重建 SR

• Image Super-Resolution 图像超分辨率重建

- SRGAN
- ESRGAN

[18] Ledig C, Theis L, Huszár F, et al.
Photo-realistic single image super-resolution using a generative adversarial network[C]//Proceedings of the IEEE conference on computer vision and pattern recognition. 2017: 4681-4690.



图像超分辨率重建 SR

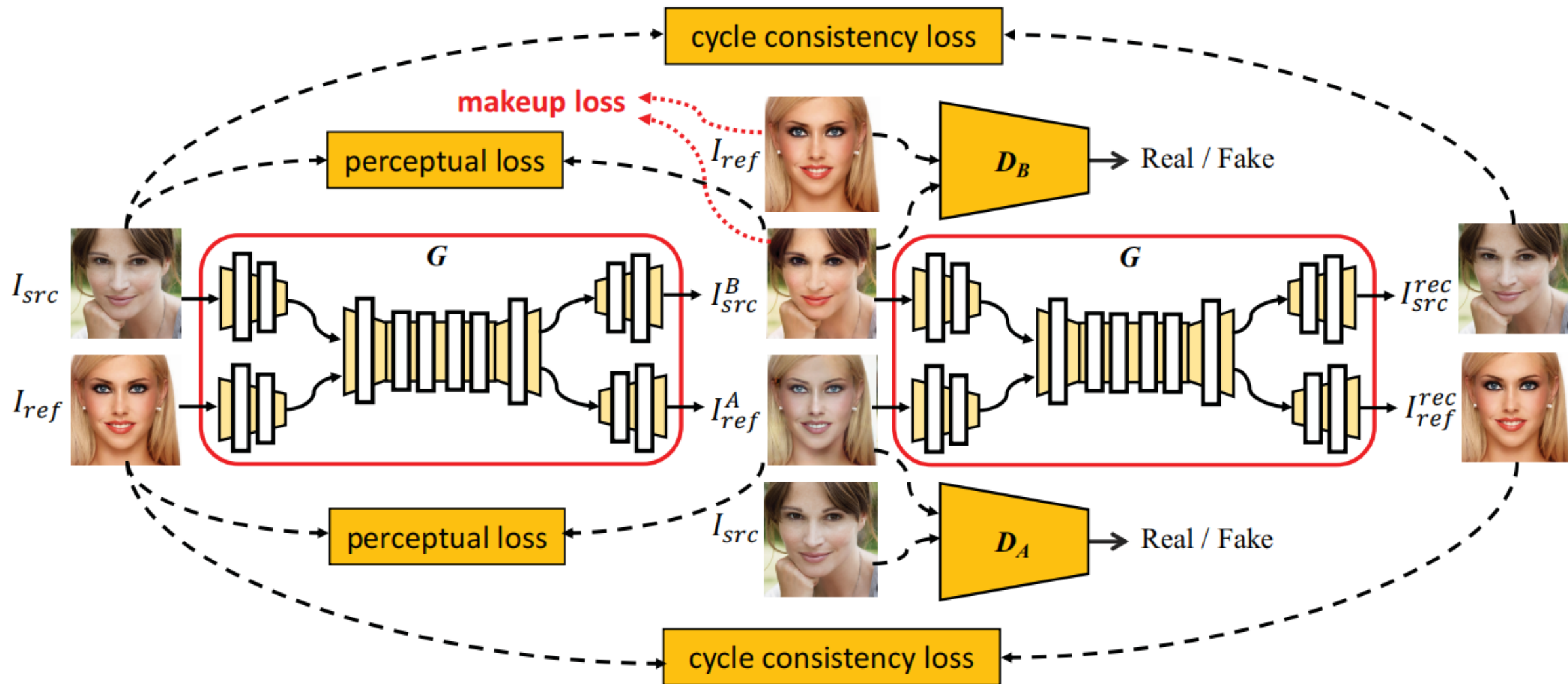
- **Anime4K:** <https://github.com/bloc97/Anime4K>

- 开源算法，能在动画播放中，实时把画面变成4k，延时低至3毫秒。
- 能把720p/1080p变成2160p，也能把480p变成1080p



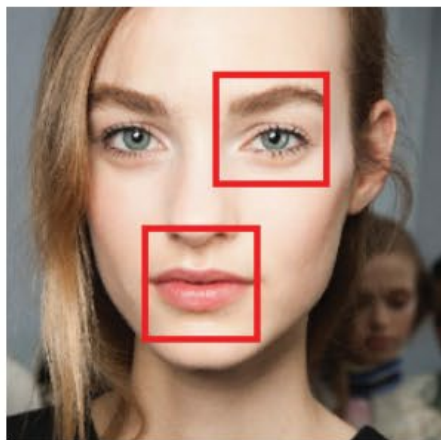
自动上妆 Makeup

• BeautyGAN

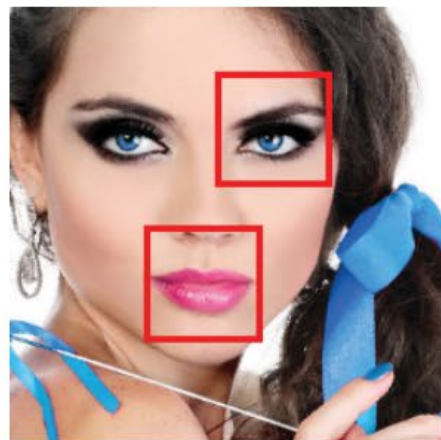
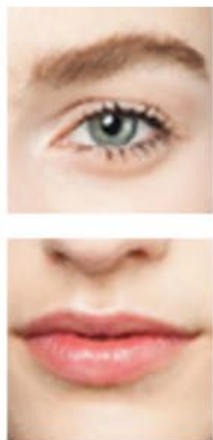


自动上妆 Makeup

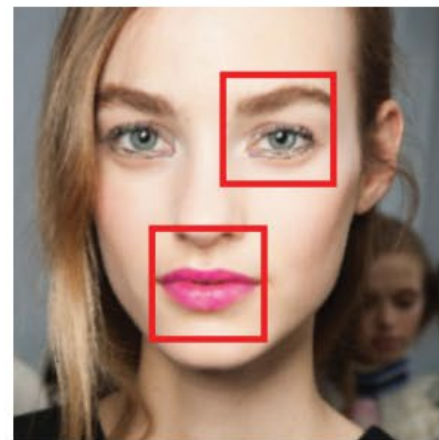
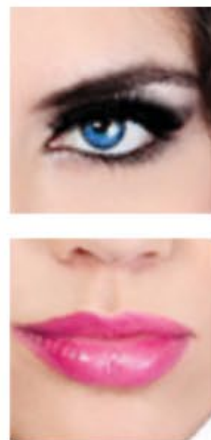
• BeautyGAN



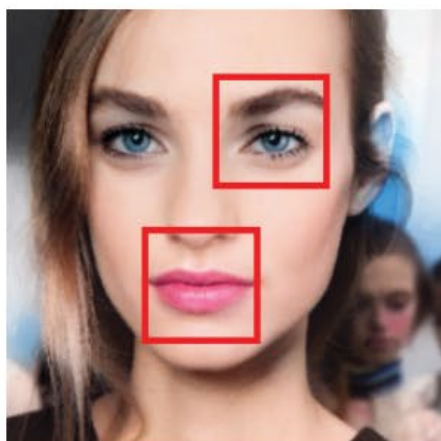
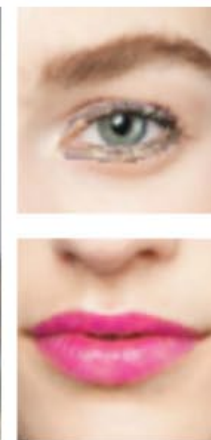
Non-makeup



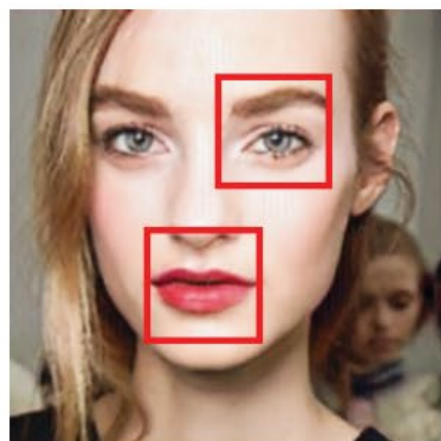
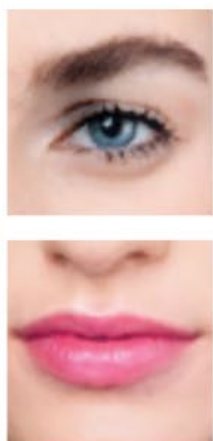
Reference



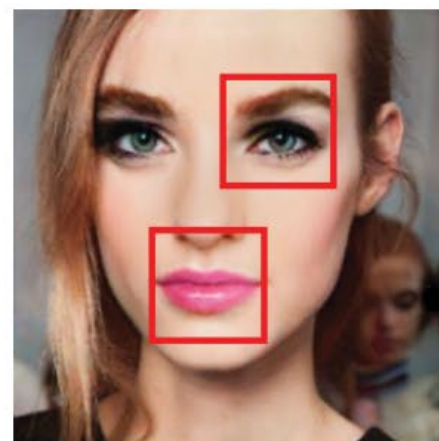
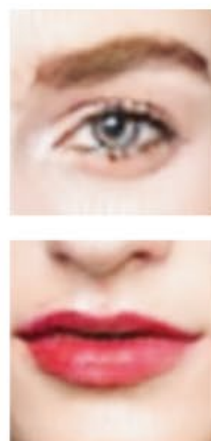
Liu[23]



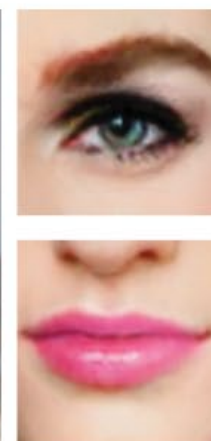
Liao[21]



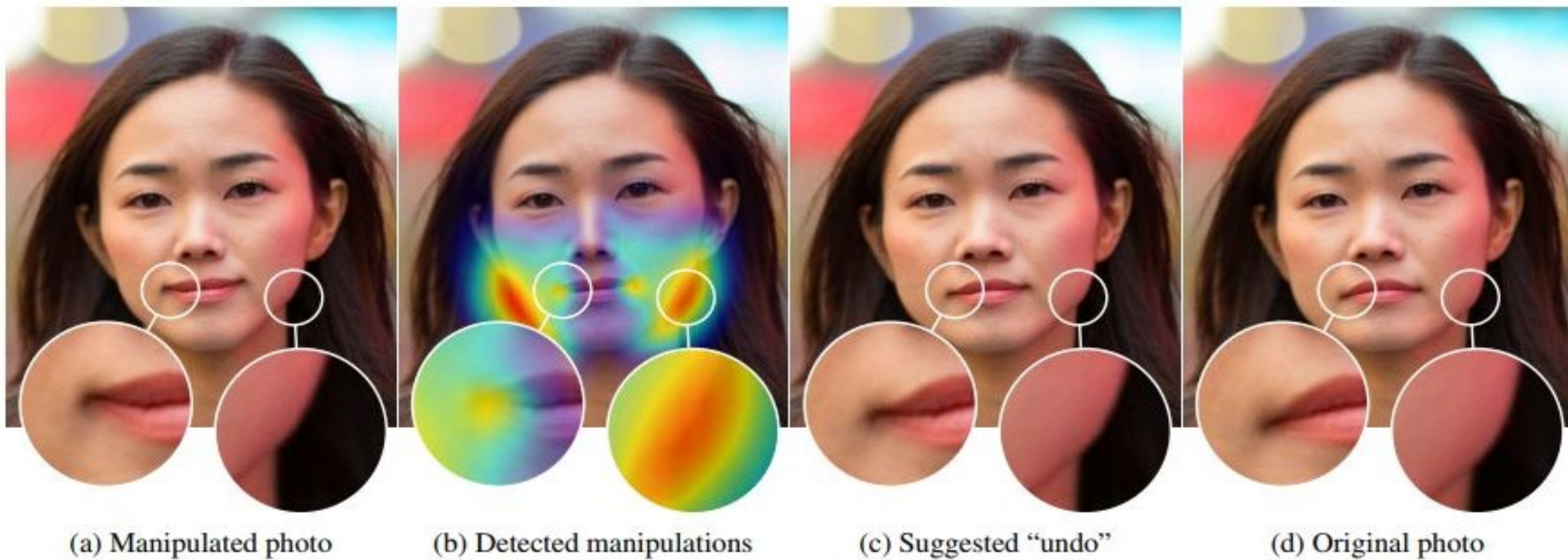
Cycle-GAN[35]



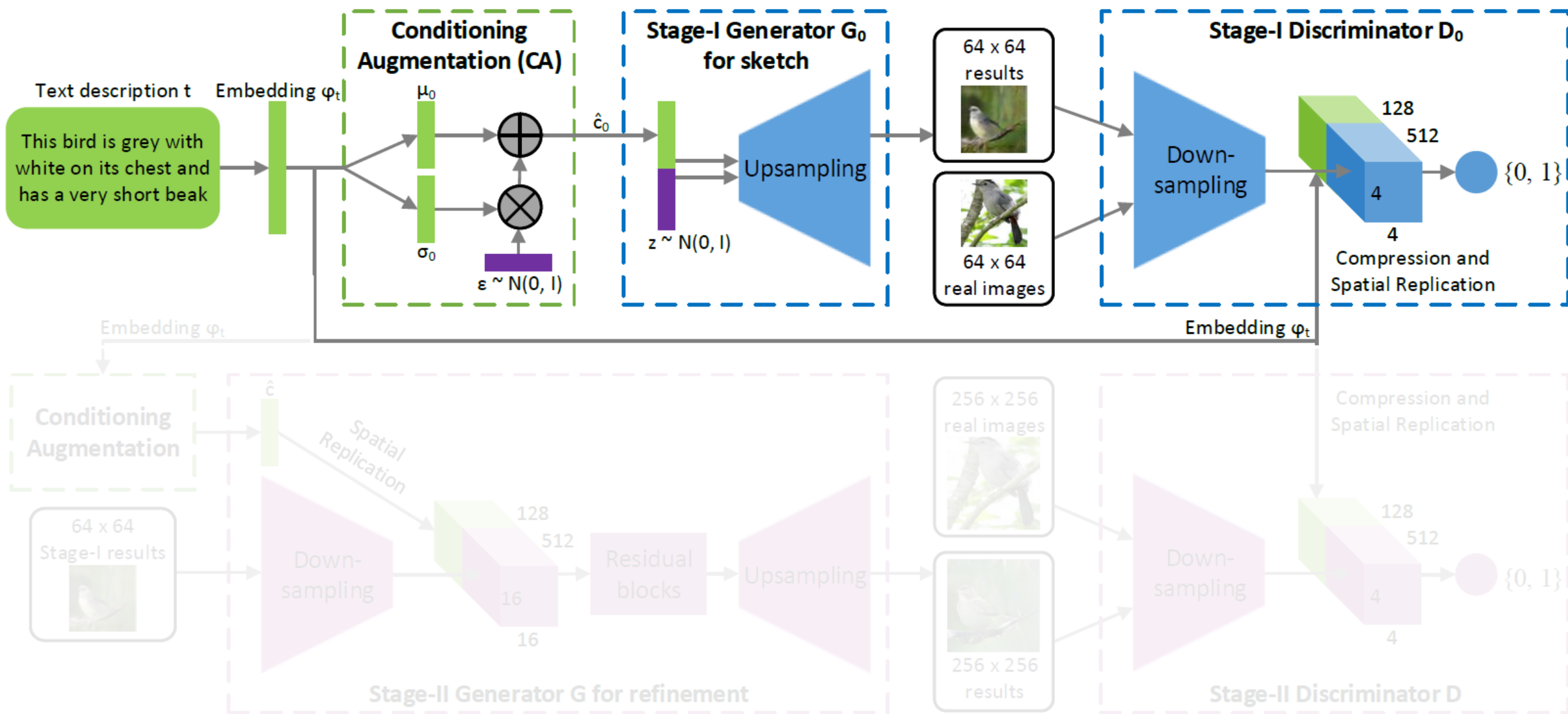
BeautyGAN



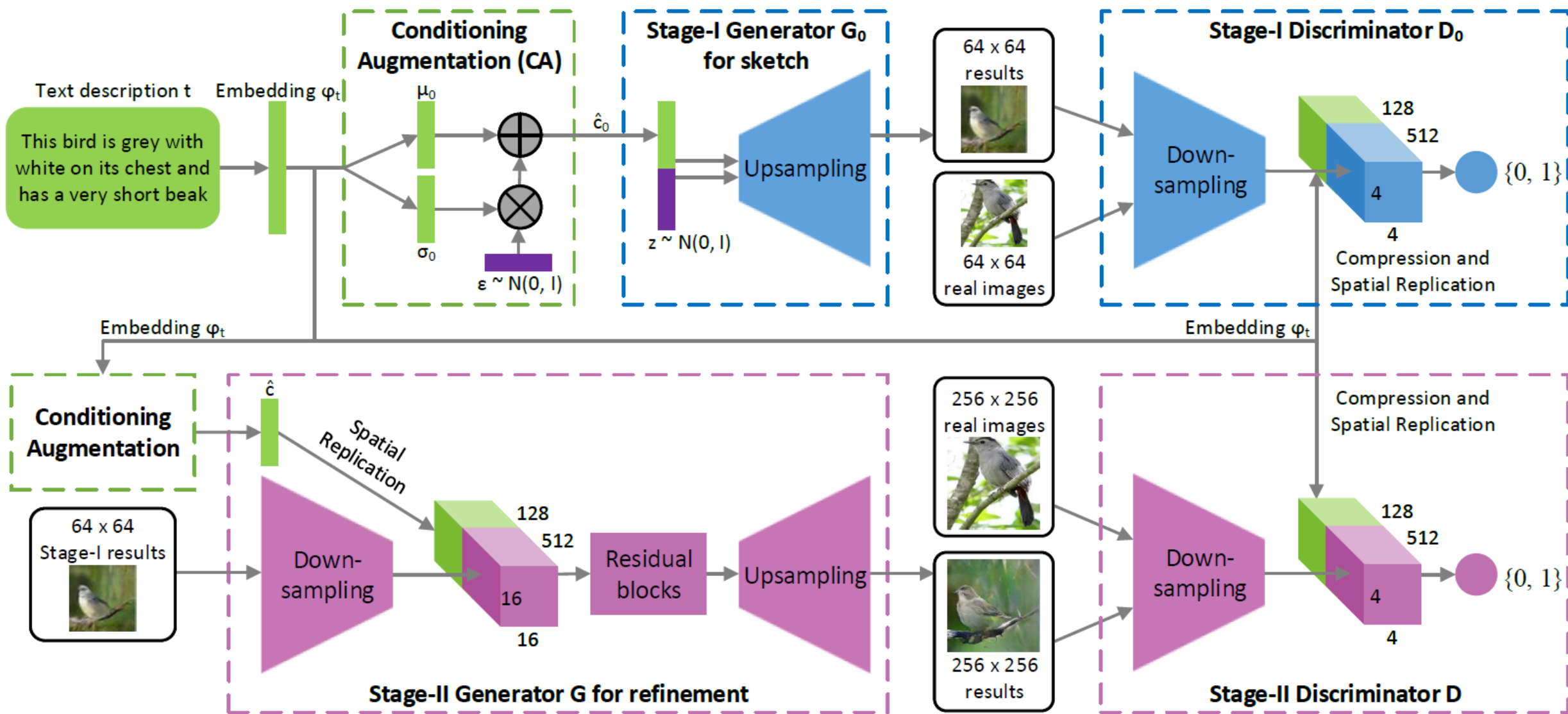
自动卸妆



文本到图像生成 Text2Image



文本到图像生成 Text2Image

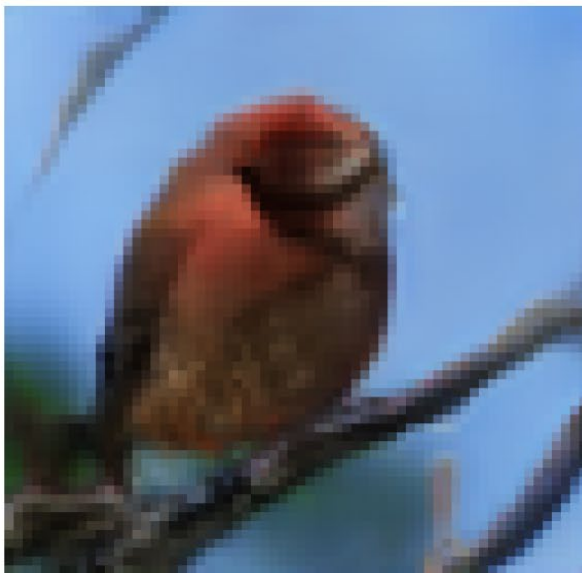


文本到图像生成

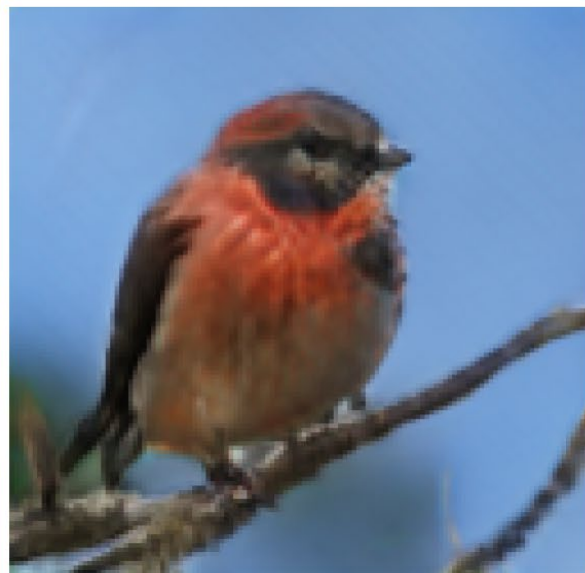
• Text2Image

- AttnGAN
- StackGAN
- StackGAN++

this bird is red with white and has a very short beak



3:red



5:white



1:bird

10:short



人脸年龄合成

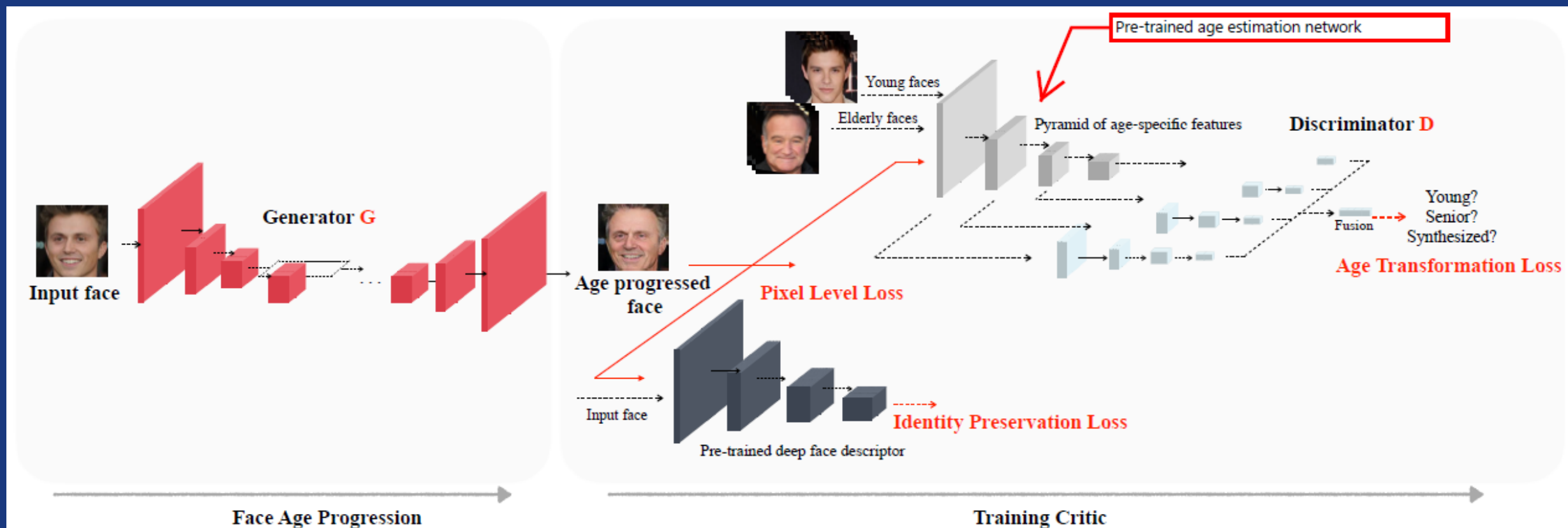
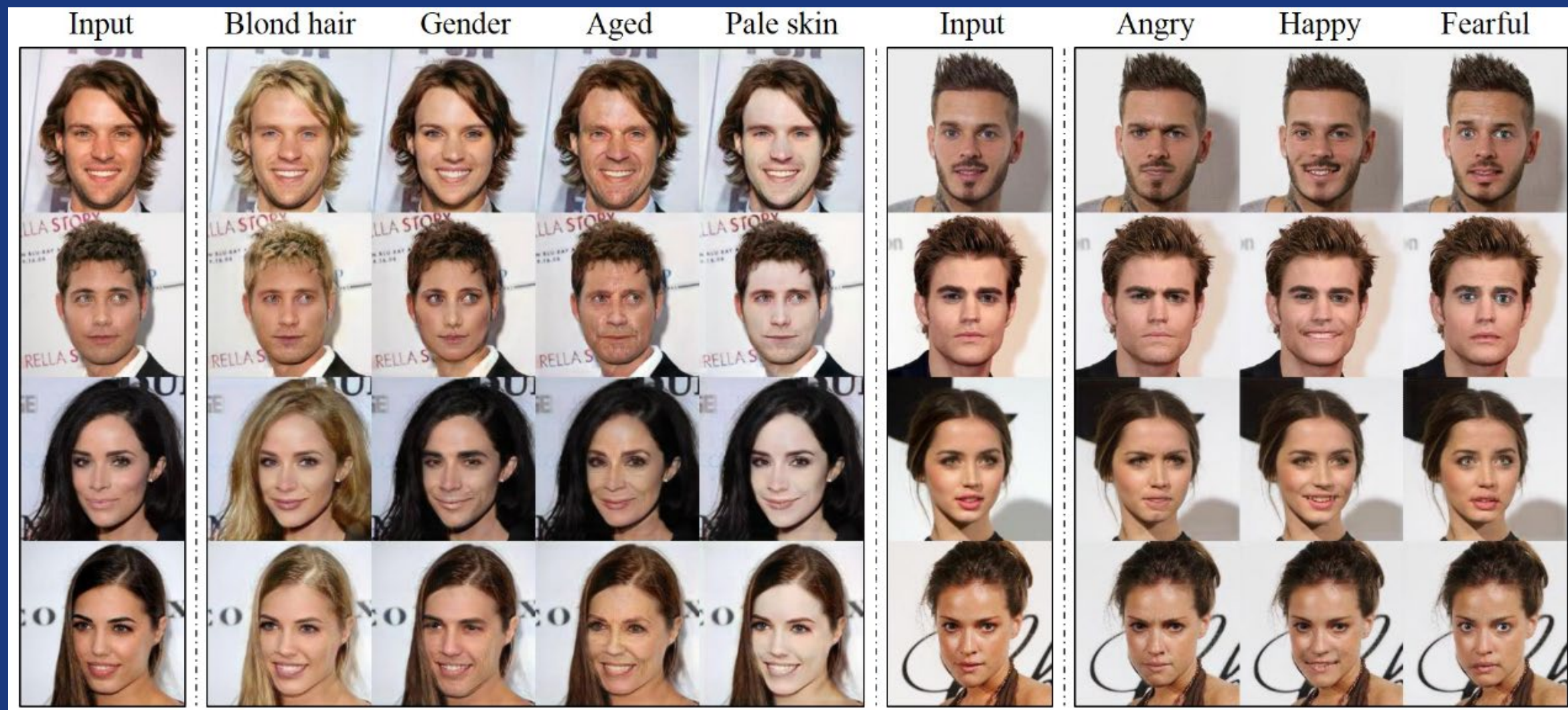


Figure 2. Framework of the proposed age progression method. A CNN based generator G learns the age transformation. The training critic incorporates the squared Euclidean loss in the image space, the GAN loss that encourages generated faces to be indistinguishable from the training elderly faces in terms of age, and the identity preservation loss minimizing the input-output distance in a high-level feature representation which embeds the personalized characteristics.

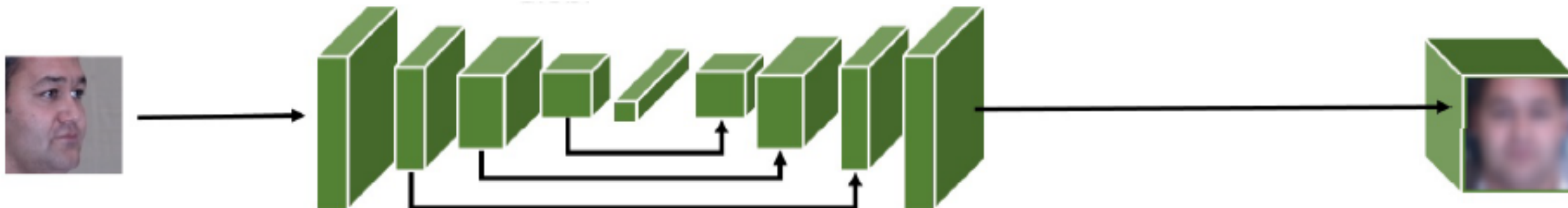
人脸属性编辑



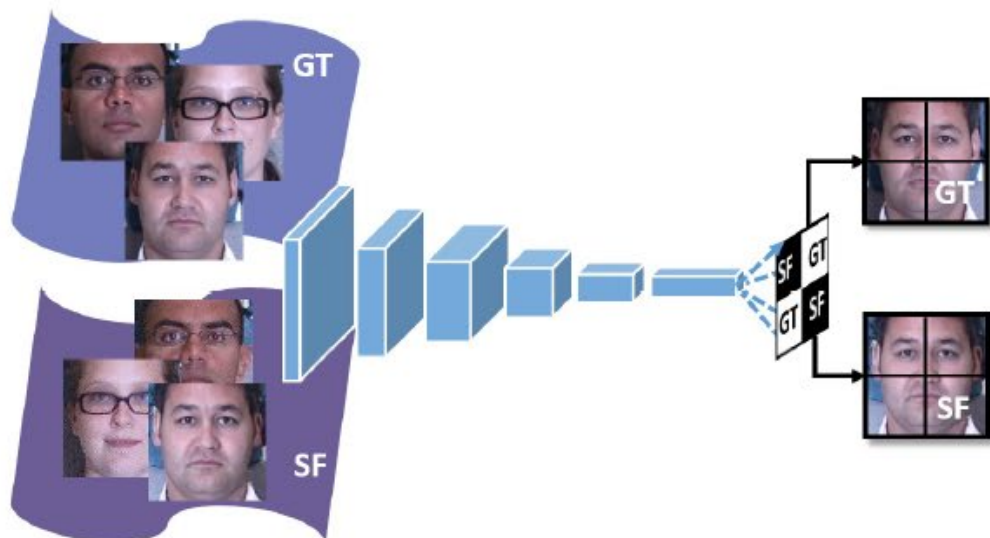
StarGAN

人脸转正

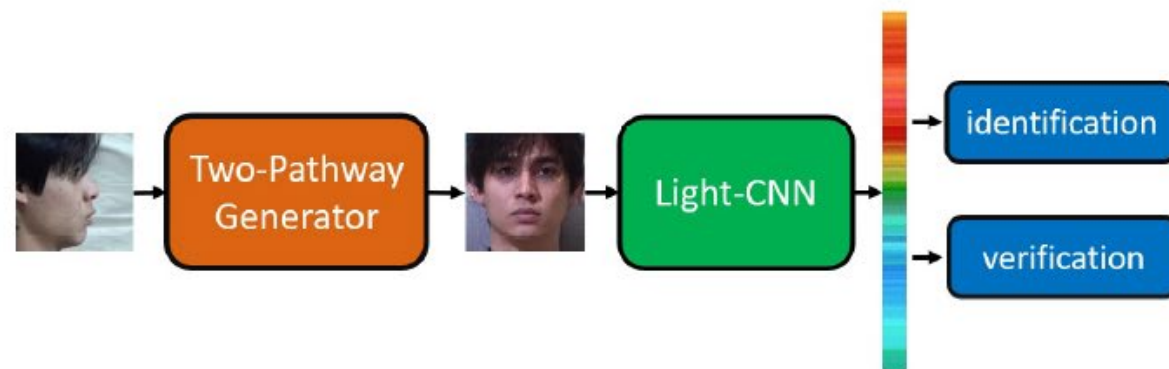
Generator



Discriminator Network

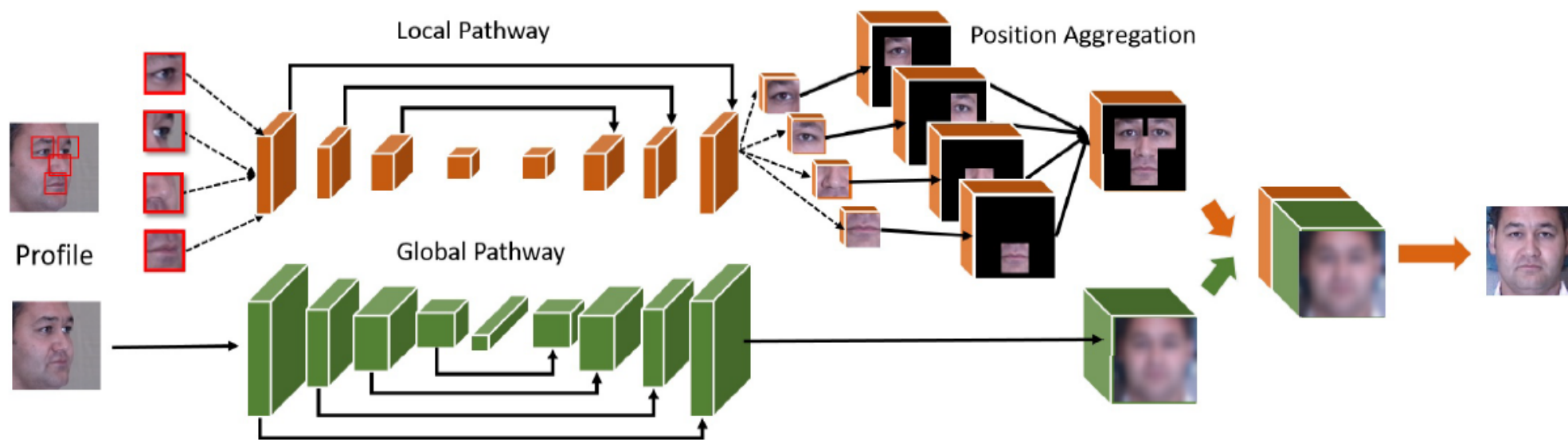


Recognition via Generation

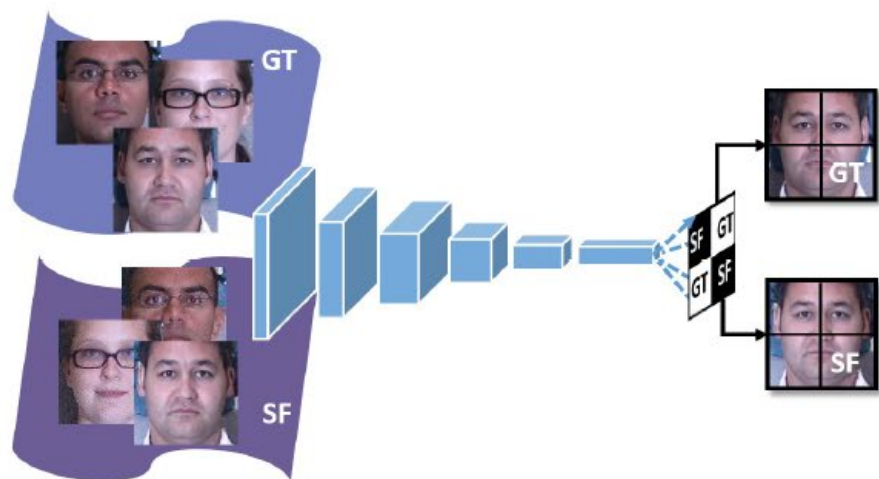


人脸转正

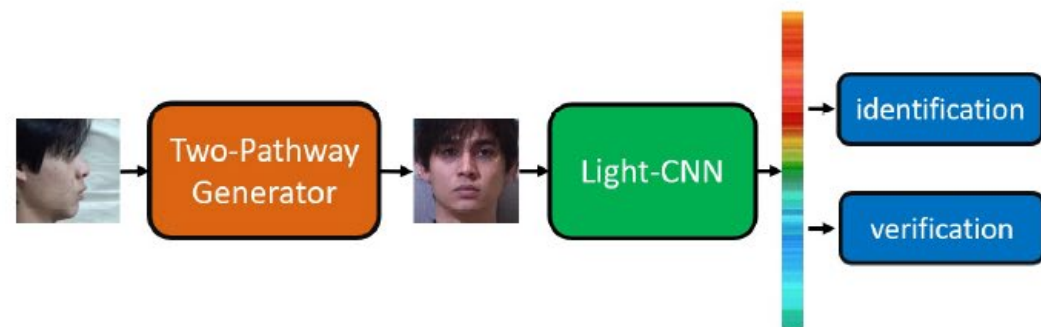
Two-pathway Generator Network



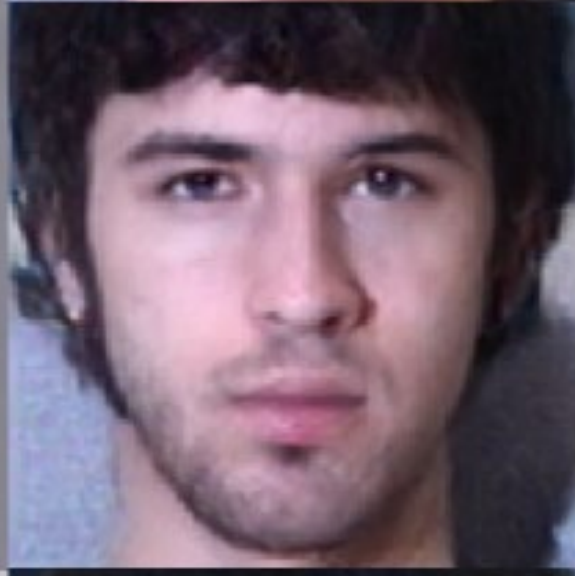
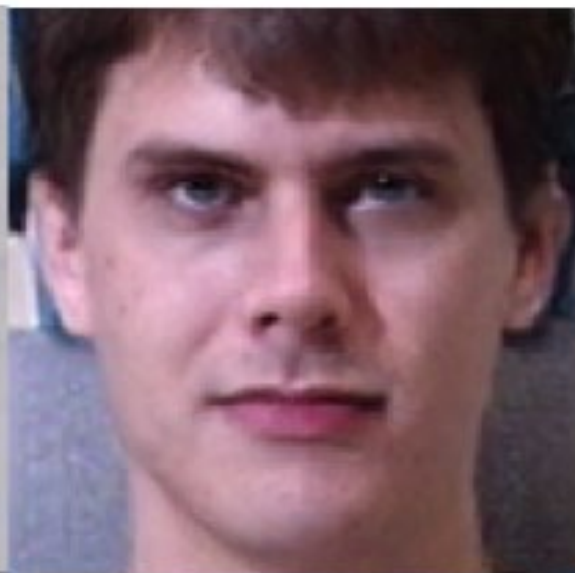
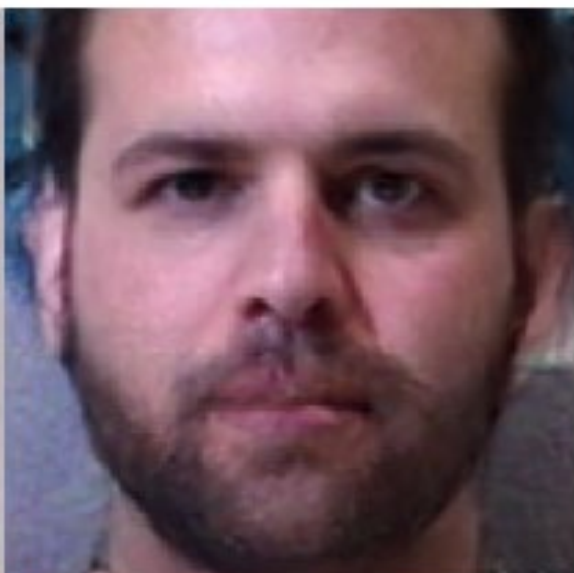
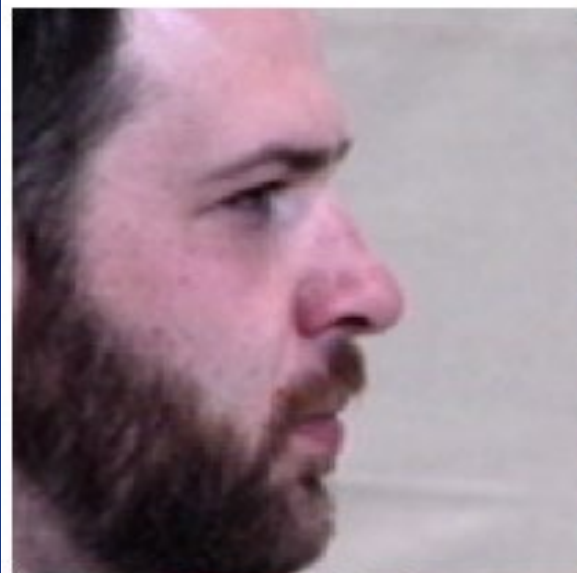
Discriminator Network



Recognition via Generation



人脸转正



图像修复

• 图像修复

- 图像去雨/去雾
- 图像补全
- 图像超分辨率重建



图像补全 Image Completion/Inpainting

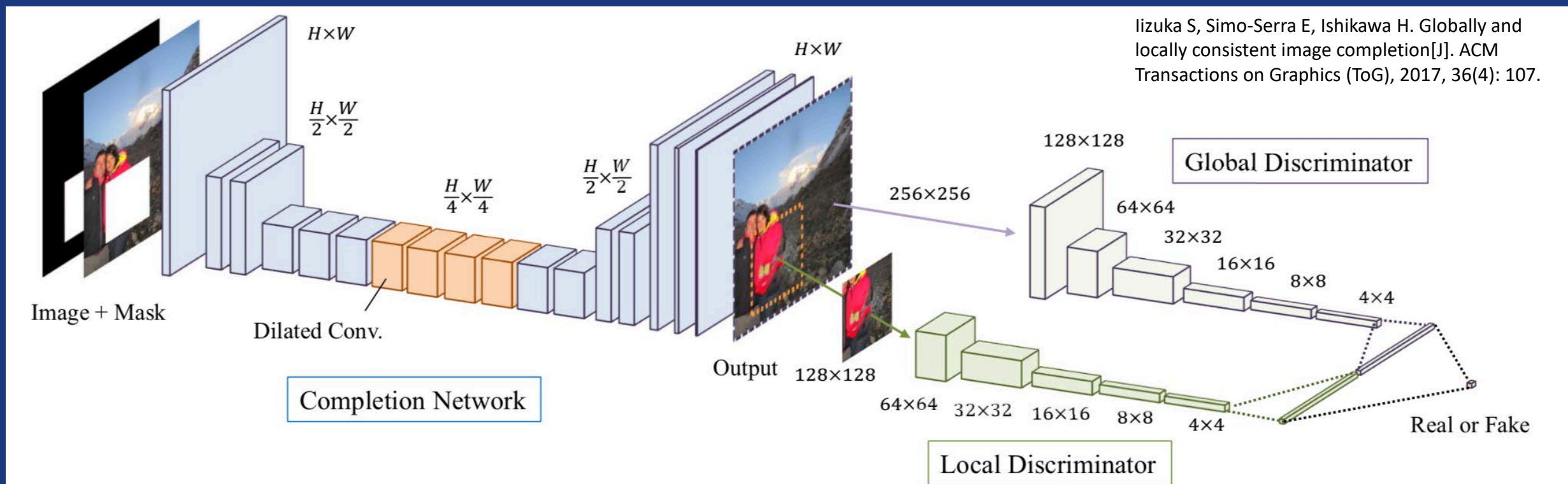
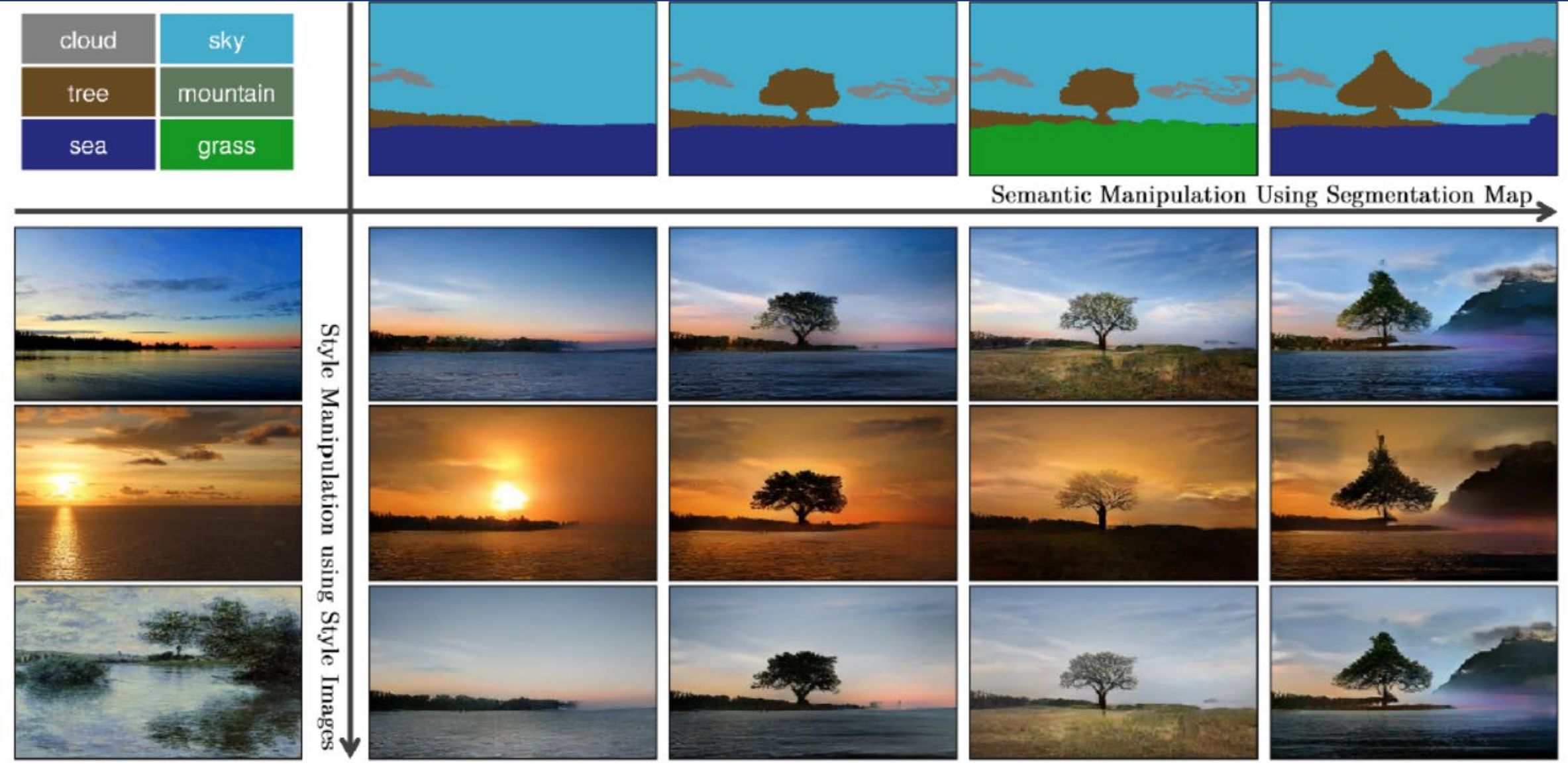


Fig. 2. Overview of our architecture for learning image completion. It consists of a completion network and two auxiliary context discriminator networks that are used only for training the completion network and are not used during the testing. The global discriminator network takes the entire image as input, while the local discriminator network takes only a small region around the completed area as input. Both discriminator networks are trained to determine if an image is real or completed by the completion network, while the completion network is trained to fool both discriminator networks.

图像合成 SPADE(GauGAN)/StyleGAN



由左至右依次为：经修改的图像、检测到修改、自动还原后的图像、原始图像

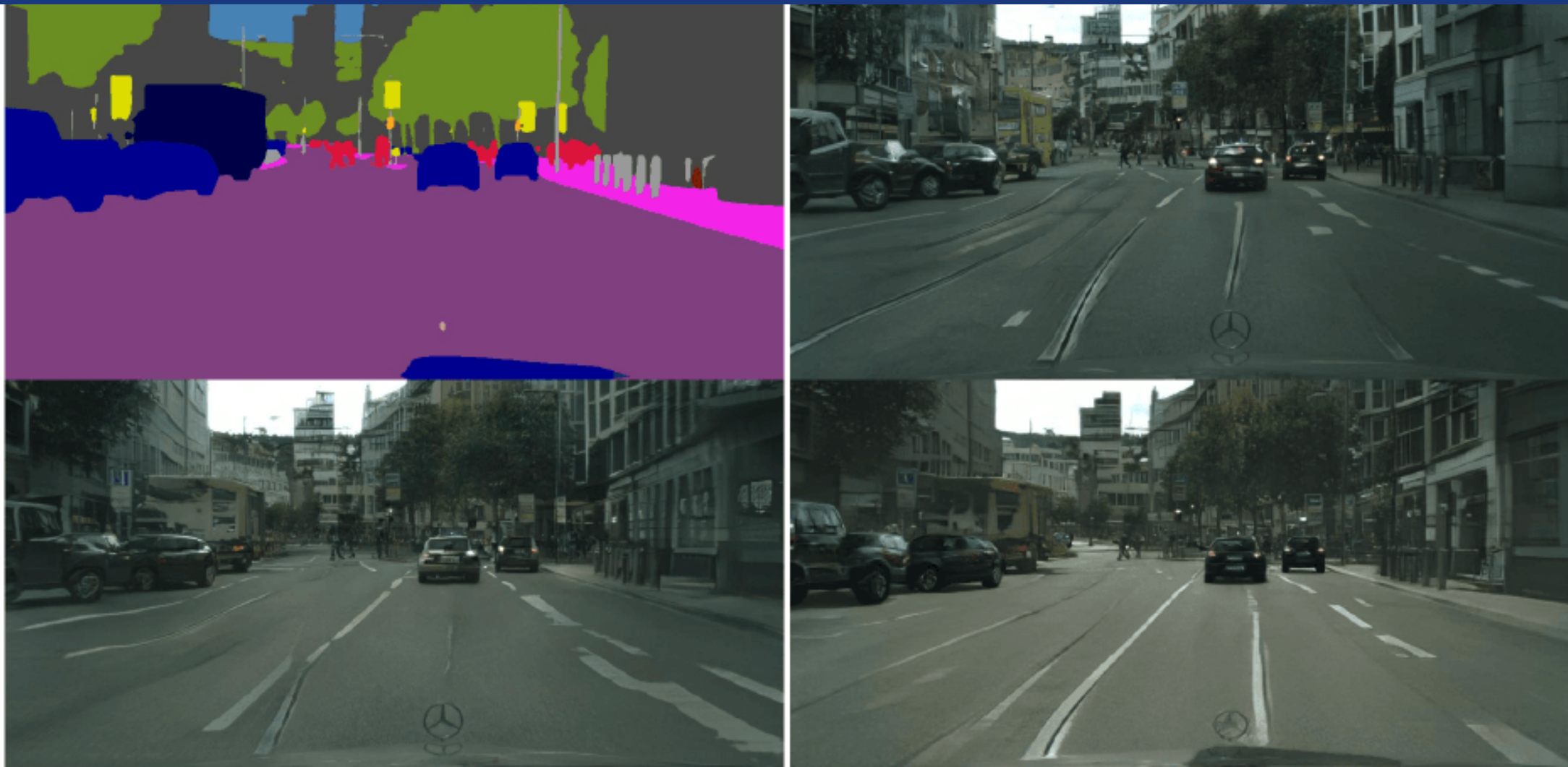


Figure 1: Generating a photorealistic video from an input segmentation map video on Cityscapes. Top left: input. Top right: pix2pixHD. Bottom left: COVST. Bottom right: vid2vid (ours). *The figure is best viewed with Acrobat Reader. Click the image to play the video clip.*

Unpaired GAN

Paired

x_i

y_i



Unpaired

X

Y



Unpaired Image-to-Image Translation using Cycle-Consistent Adversarial Networks

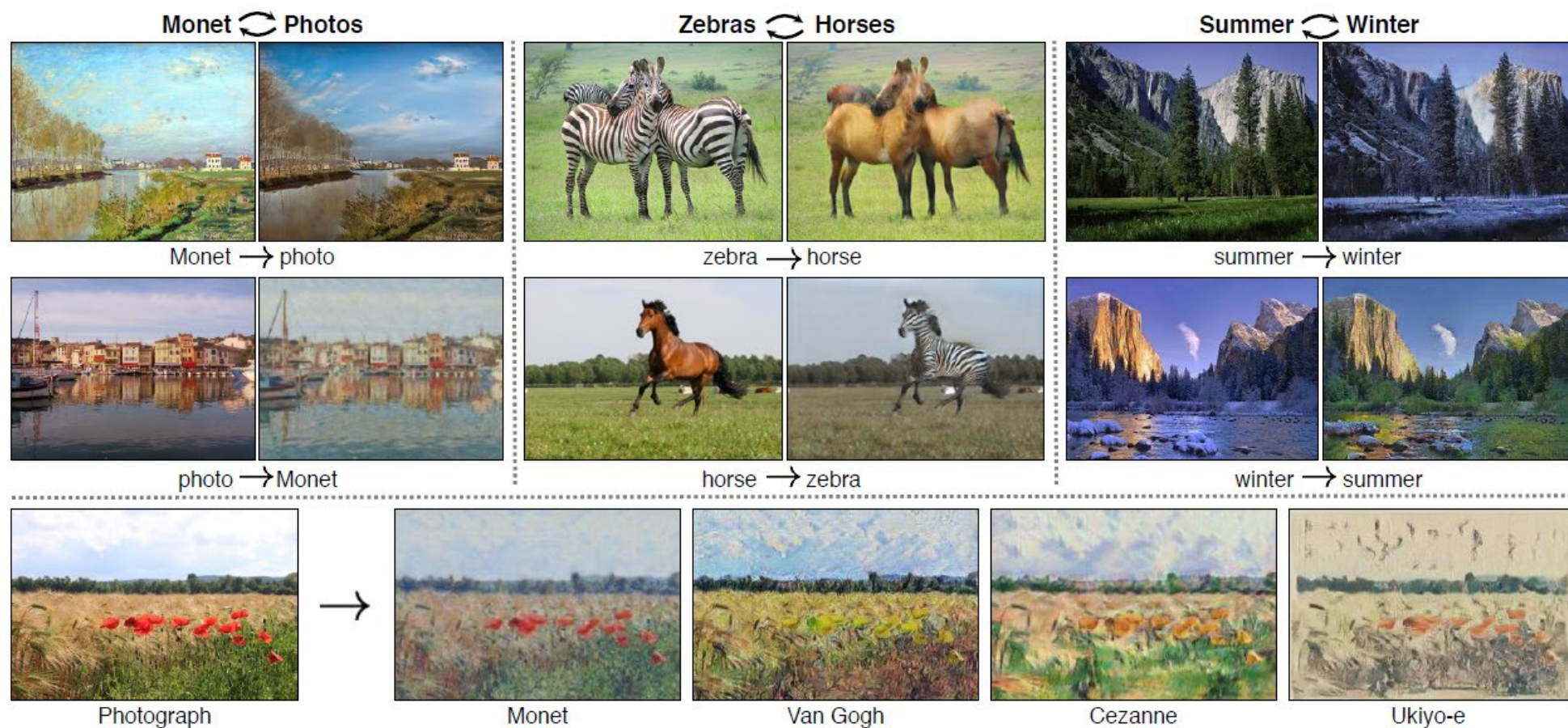
Jun-Yan Zhu*

Taesung Park*

Phillip Isola

Alexei A. Efros

Berkeley AI Research (BAIR) laboratory, UC Berkeley



CycleGAN

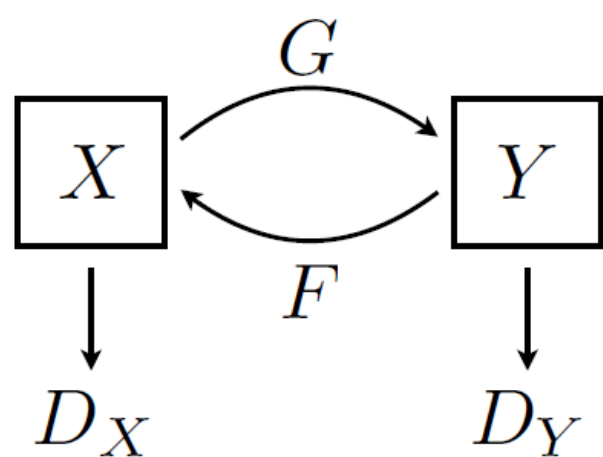
Input x



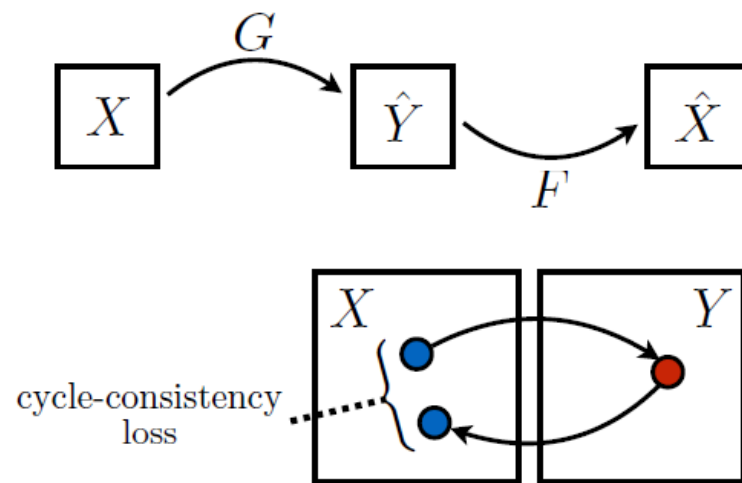
Generated image $G(x)$



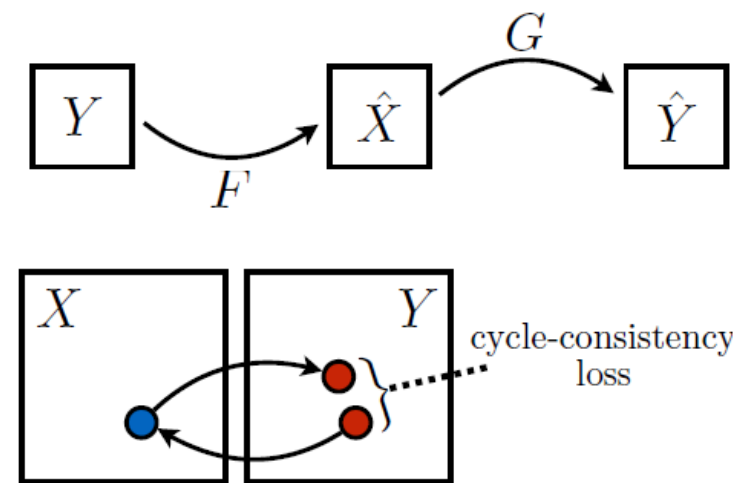
Reconstruction $F(G(x))$



(a)

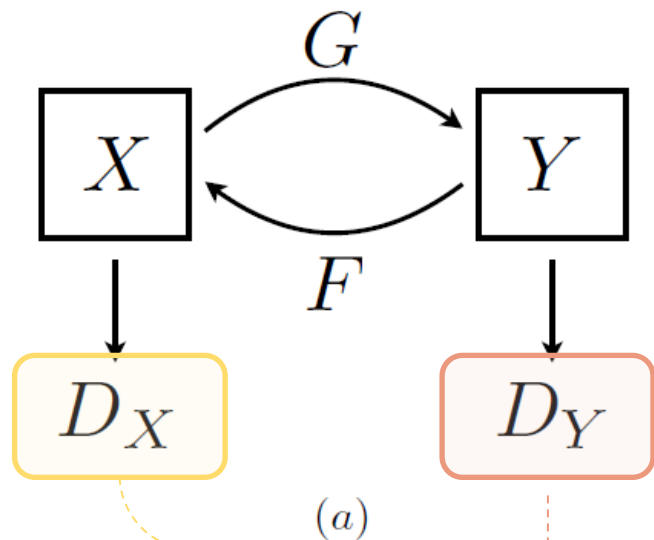


(b)



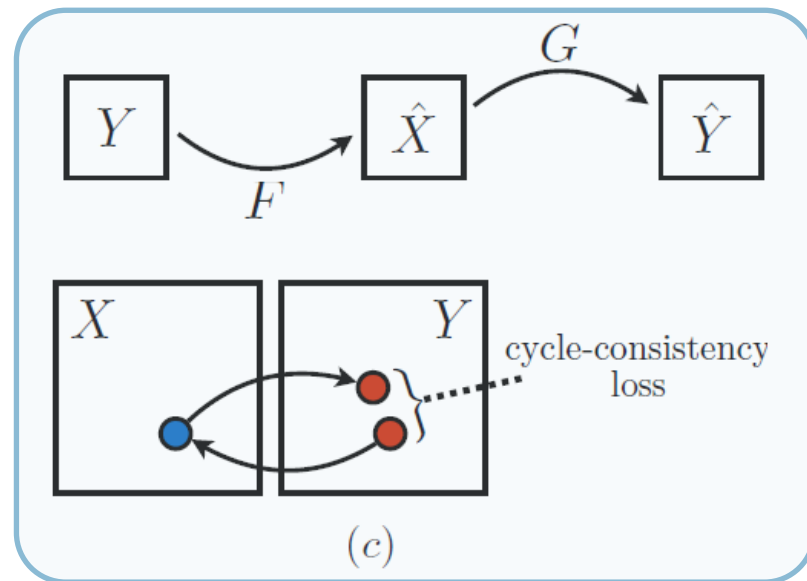
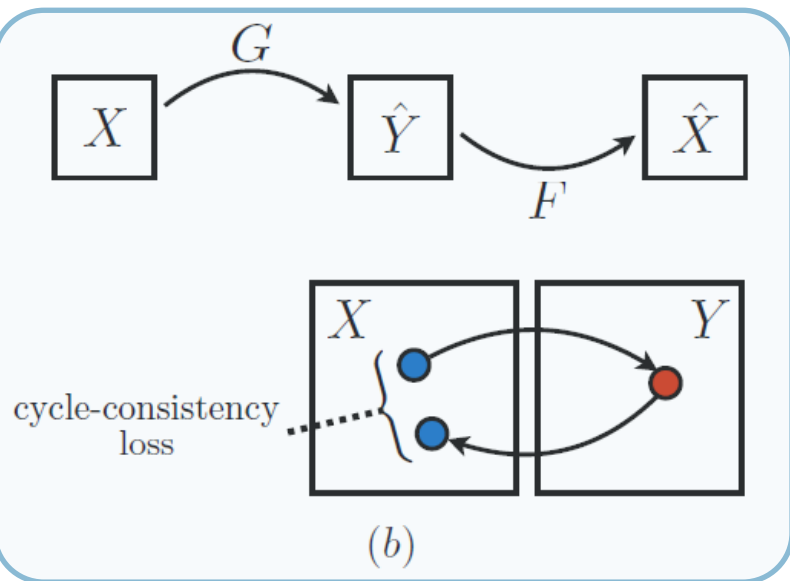
(c)

CycleGAN



$$\mathcal{L}_{\text{GAN}}(F, D_X, Y, X)$$

$$\mathcal{L}_{\text{GAN}}(G, D_Y, X, Y) = \mathbb{E}_{y \sim p_{\text{data}}(y)} [\log D_Y(y)] + \mathbb{E}_{x \sim p_{\text{data}}(x)} [\log(1 - D_Y(G(x)))]$$



$$\mathcal{L}_{\text{cyc}}(G, F) = \mathbb{E}_{x \sim p_{\text{data}}(x)} [\|F(G(x)) - x\|_1] + \mathbb{E}_{y \sim p_{\text{data}}(y)} [\|G(F(y)) - y\|_1]$$

CycleGAN

Input



Monet



Van Gogh



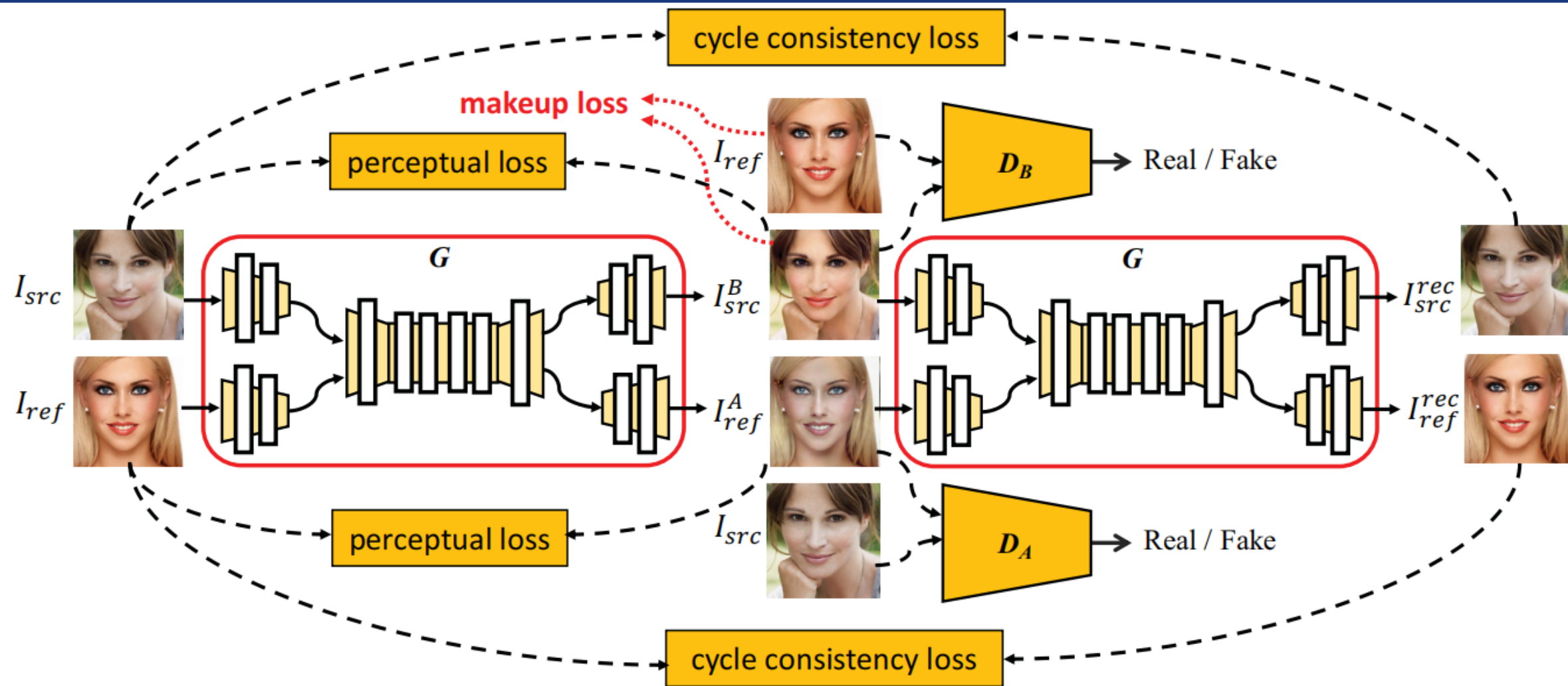
Cezanne



Ukiyo-e



自动上妆 Makeup: BeautyGAN、PSGAN



Q&A